

IS328: Data Mining

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Assignment 1

Group 4

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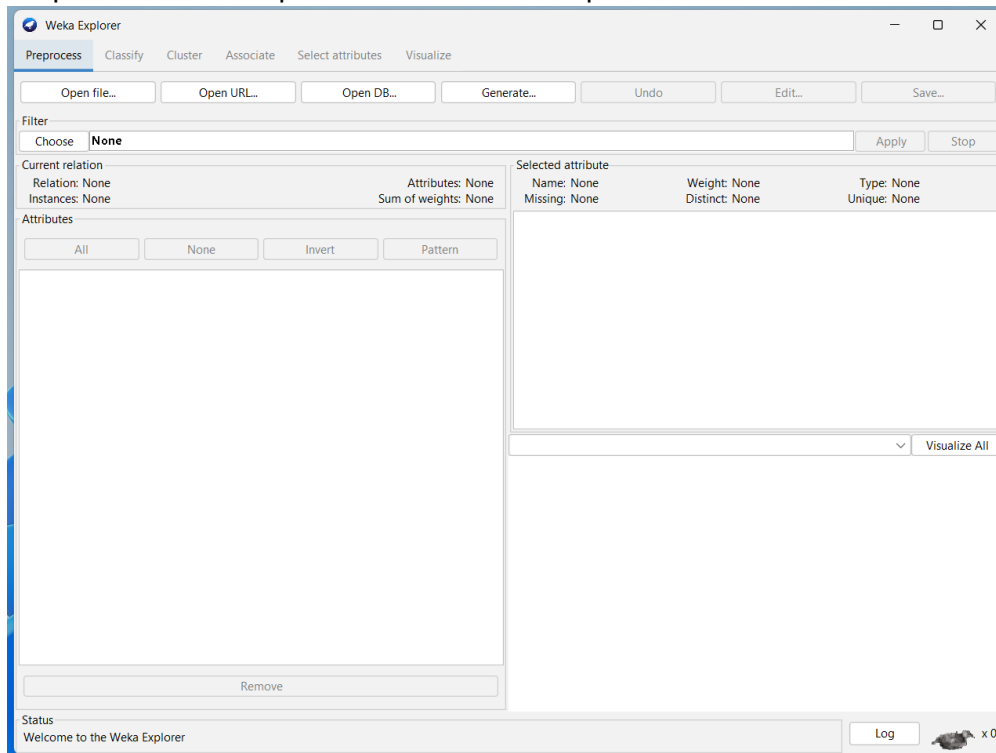
1.0 Data Pre-processing

1.1 Attribute Selection and Justification:

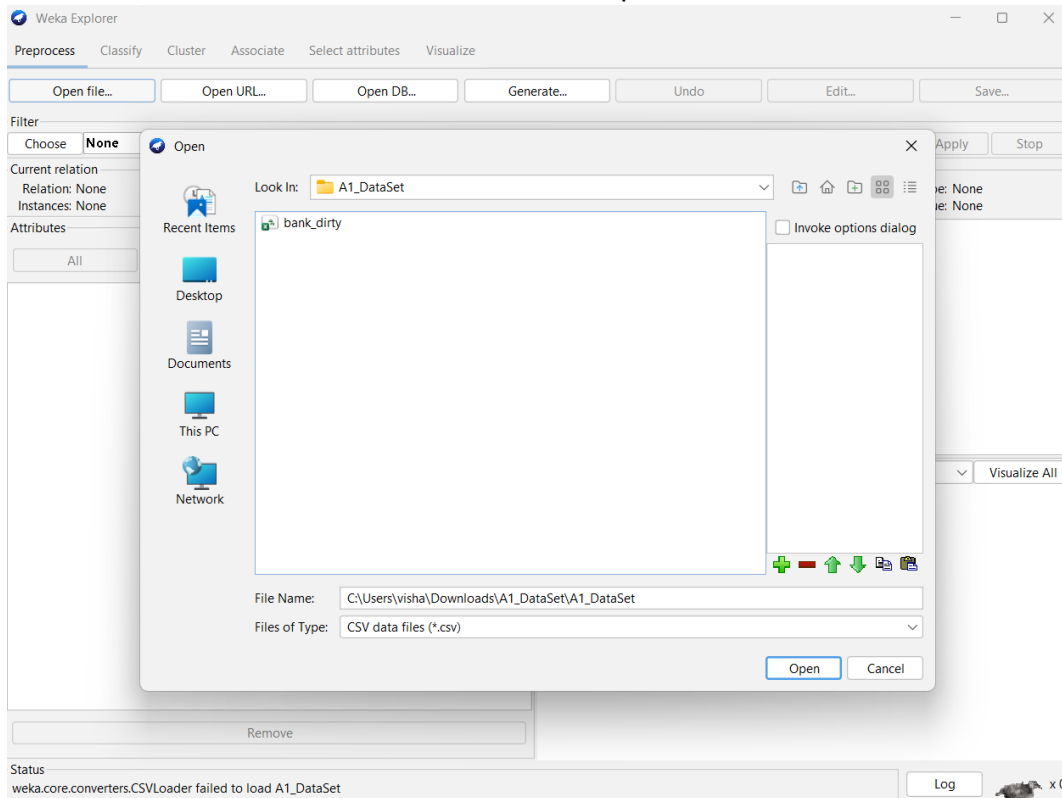
Step 1 – Open the WEKA Application and click on “Explorer” under “Applications”.



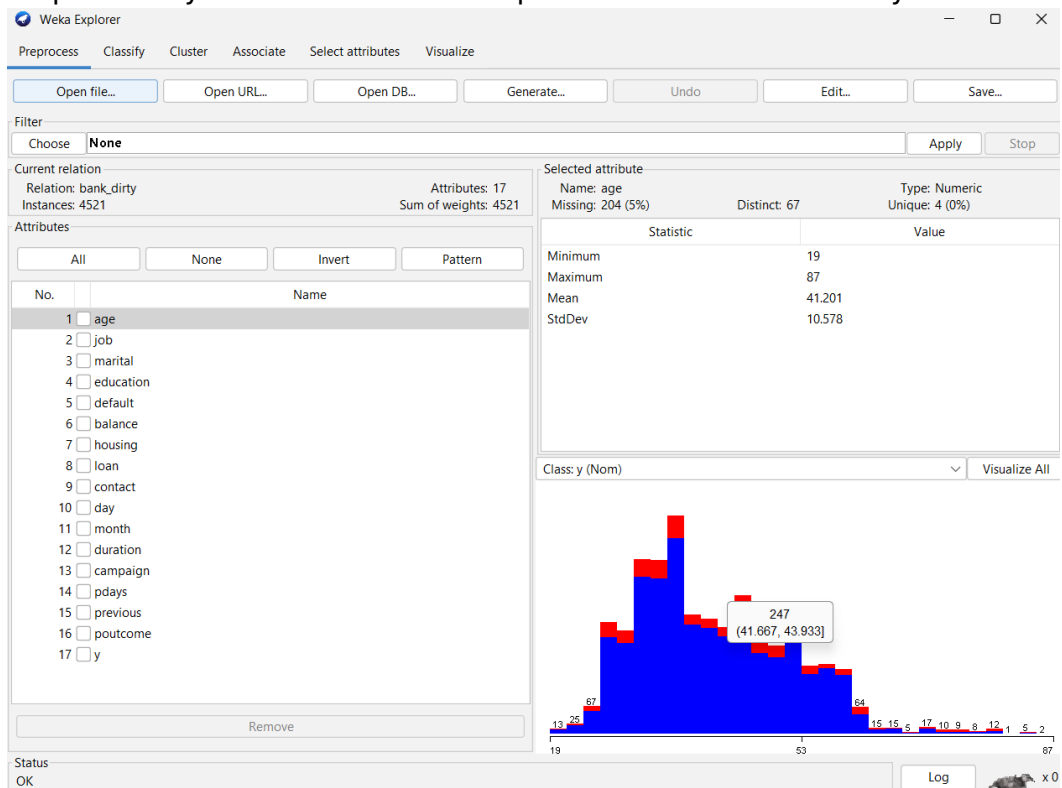
Step 2 – Click on “Open file” under the Preprocess tab.



Step 3 – Navigate to the location of the “bank_dirty.csv” file, set the “Files of Type” to CSV data files, select the data file and click on “Open”.



Step 4 – Verify that the data file was opened in WEKA successfully.



Step 5 – In order aid in determining the redundant and irrelevant attributes that should be removed, two types of attribute evaluators were used. These consisted of the Information Gain Attribute Evaluator and Correlation Attribute Evaluator. In order to select the attribute evaluators, click on the “Select attributes” tab and click on the “Choose” button under “Attribute Evaluator”, select the attribute evaluators of your choice and choose the appropriate search methods, in this case, we have decided to use “Ranker” for both evaluators.

Result of Information Gain Attribute Evaluator:

The screenshot shows the Weka Explorer interface with the 'Select attributes' tab active. The 'Attribute Evaluator' is set to 'InfoGainAttributeEval' and the 'Search Method' is 'Ranker -T -1.7976931348623157E308 -N -1'. The 'Attribute Selection Mode' is 'Use full training set'. The 'Attribute selection output' pane displays the following results:

```

=== Attribute Selection on all input data ===

Search Method:
  Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 17 y):
  Information Gain Ranking Filter

Ranked attributes:
0.10811967 12 duration
0.03553361 14 pdays
0.03204198 16 poutcome
0.0299014 11 month
0.01622639 15 previous
0.0138146 9 contact
0.00927256 1 age
0.0083454 2 job
0.00782731 7 housing
0.00553494 6 balance
0.0041129 8 loan
0.00304631 13 campaign
0.00297254 3 marital
0.00226067 4 education
0.00000121 5 default
0 10 day

Selected attributes: 12,14,16,11,15,9,1,2,7,6,8,13,3,4,5,10 : 16
  
```

The 'Result list (right-click for options)' pane shows a single entry: '13:00:35 - Ranker + InfoGainAttributeEval'.

Results of Correlation Attribute Evaluator:

The screenshot shows the Weka Explorer interface with the 'Attribute Evaluator' tab selected. The 'Attribute Evaluator' dropdown is set to 'CorrelationAttributeEval'. The 'Search Method' is 'Ranker -T -1.7976931348623157E308 -N -1'. The 'Attribute Selection Mode' is 'Use full training set'. The 'Result list' on the left shows three entries, with '13:08:31 - Ranker + CorrelationAttributeEval' selected. The 'Attribute selection output' pane on the right displays the following text:

```

=== Attribute Selection on all input data ===

Search Method:
  Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 17 y):
  Correlation Ranking Filter

Ranked attributes:
0.40112 12 duration
0.13986 16 poutcome
0.11671 15 previous
0.11164 9 contact
0.10468 7 housing
0.10409 14 pdays
0.07052 8 loan
0.06115 13 campaign
0.05618 3 marital
0.0531 11 month
0.0453 1 age
0.03155 2 job
0.02967 4 education
0.0237 6 balance
0.01124 10 day
0.0013 5 default

Selected attributes: 12,16,15,9,7,14,8,13,3,11,1,2,4,6,10,5 : 16
  
```

Expected Outputs:

Upon evaluating each attribute via the information gain attribute evaluator and the correlation attribute evaluator, the following attributes are selected to be removed:

- i. **Default** – The attribute “default” produces a significantly low information gain value of 0.00000121, indicating that it has a very low predicting power in terms of determining the target class, which is whether a client will subscribe to a term deposit (y = yes/no). The attribute “default” also produces the lowest correlation value of 0.0013, out of all the other attributes, indicating a weak relationship between the attribute, “default” and the target class (y = yes/no).
- ii. **Day** – The attribute “day” produces an information gain value of 0, indicating that it has no predicting power at all in terms of determining the target class (y = yes/no). The attribute “day” also produces a significantly low correlation value of 0.01124, which outlines that this attribute has a very weak relationship with the target class (y = yes/no) and it has no impact on the target class.
- iii. **Previous** – The attribute “previous” produces an information gain value of 0.01622639 and a correlation value of 0.11671. While these values indicate moderate information gain and correlation of attribute “previous”, the attribute consists of high number values as 0s. This indicates high sparsity, and it also redundancy.

- iv. **Pdays** – The attribute “pdays” produces an information gain value of 0.03553361 and a correlation value of 0.10409. While these values indicate that the attribute “pdays” has a moderate relationship with the target class (y = yes/no), the attribute consists of a high proportion of values as -1. This indicates that clients were not contacted earlier on, and it also indicates redundancy.

The list of attributes that were kept are age, job, marital, education, balance, housing, loan, contact, month, duration, campaign and poutcome. These attributes had produced significant results of information gain and correlation which indicated that these 12 attributes had a significant impact on the class (y = yes/no).

Step 6 – In order to remove the chosen attributes, select the “Remove” filter from the filter list.

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Open file... | Open URL... | Open DB... | Generate... | Undo | Edit... | Save...

Filter: Choose | **Remove** | Apply | Stop

Current relation
Relation: bank_dirty
Instances: 4521
Attributes: 17
Sum of weights: 4521

Attributes

All | None | Invert | Pattern

No.	Name
1	age
2	job
3	marital
4	education
5	default
6	balance
7	housing
8	loan
9	contact
10	day
11	month
12	duration
13	campaign
14	pdays
15	previous
16	poutcome
17	y

Remove

Selected attribute
Name: default
Missing: 0 (0%)
Distinct: 2
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	no	4445	4445
2	yes	76	76

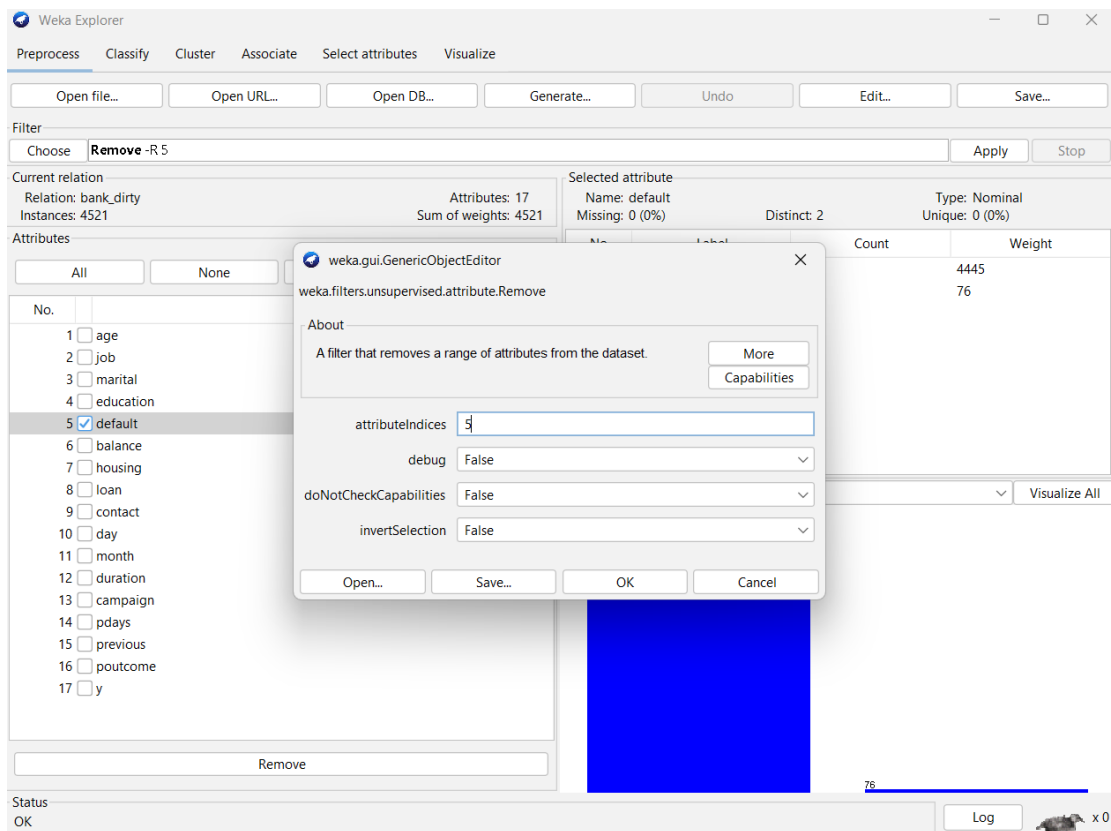
Class: y (Nom) | Visualize All

4445

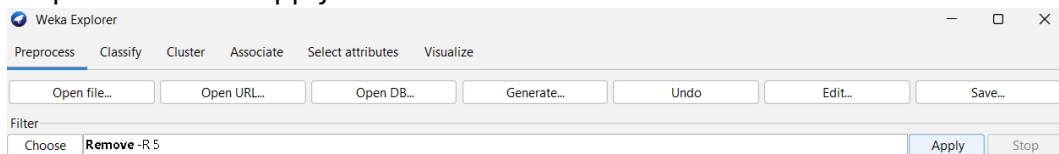
76

Status: OK | Log | x 0

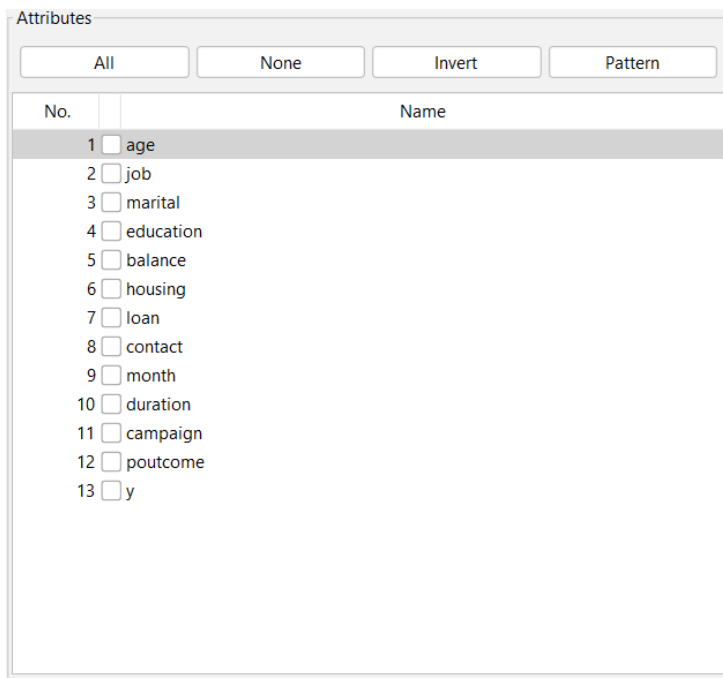
Step 7 – Click on the Remove filter and set the attribute indices of the attribute to be removed (in this case, attribute “default” is at index 5). Then click on “OK”.



Step 8 – Click on “Apply” to remove the selected attribute.

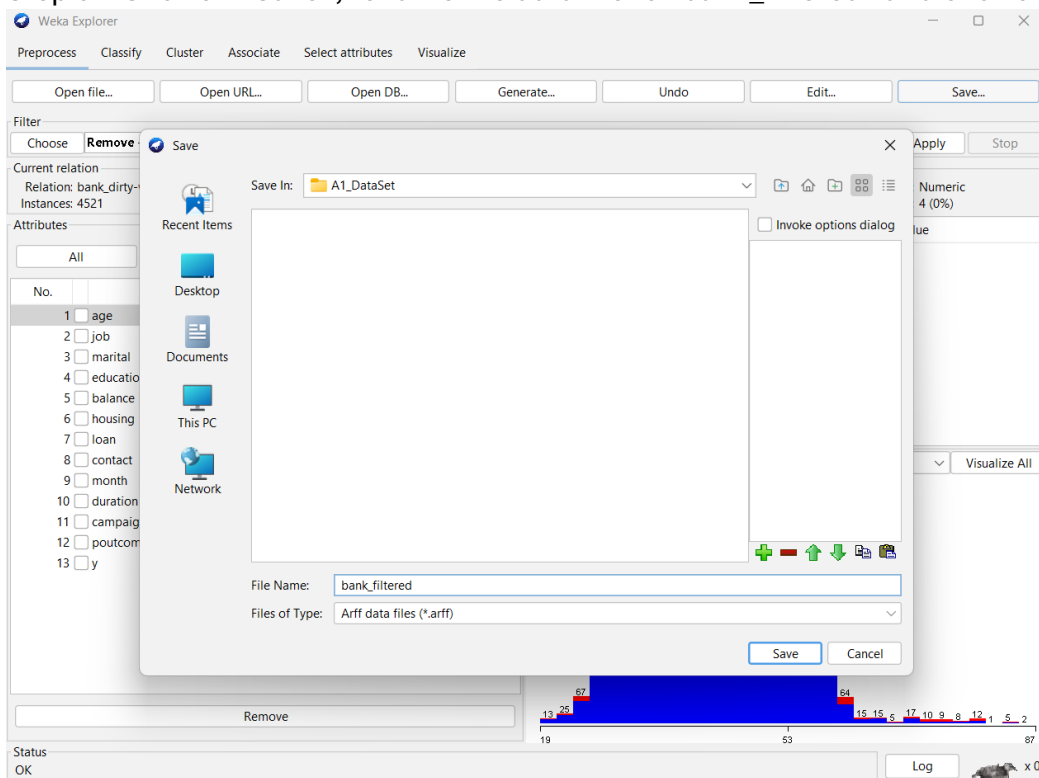


Repeat steps 7 & 8 until all the attributes selected to be removed are removed.
Expected Outputs:



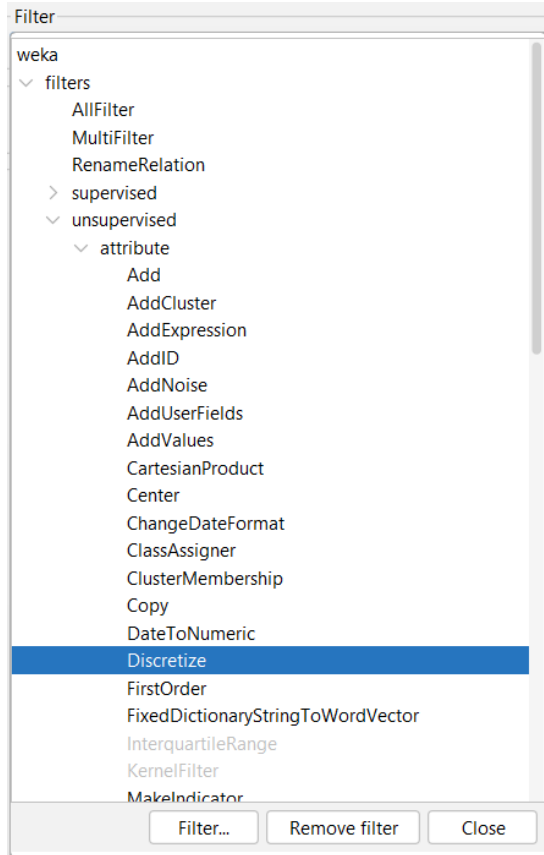
WEKA interface showing final attribute list.

Step 9 – Click on “Save”, rename the data file to “bank_filtered” and click on “Save”.



1.2 Discretization and Normalization:

Step 1 – To perform discretization, select the “Discretize” filter from the filter list.



Step 2 – Click on the Discretize filter.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter: Choose **Discretize** -B 10 -M -1.0 -R first-last -precision 6 Apply Stop

Current relation
 Relation: bank_dirty-weka.filters.unsupervised.attribute.Remove... Attributes: 13
 Instances: 4521 Sum of weights: 4521

Attributes

All None Invert Pattern

No.	Name
1	☒ age
2	☐ job
3	☐ marital
4	☐ education
5	☐ balance
6	☐ housing
7	☐ loan
8	☐ contact
9	☐ month
10	☐ duration
11	☐ campaign
12	☐ poutcome
13	☐ y

Remove

Selected attribute

Name: age		Type: Numeric
Missing: 204 (5%)	Distinct: 67	Unique: 4 (0%)
Statistic	Value	
Minimum	19	
Maximum	87	
Mean	41.201	
StdDev	10.578	

Class: y (Nom) Visualize All

Status: OK Log x 0

Step 3 – Set the attribute indices as the index of the attribute “age”, which is 1 in this case, set bins to 3 and click on “OK”.

weka.gui.GenericObjectEditor

weka.filters.unsupervised.attribute.Discretize

About

An instance filter that discretizes a range of numeric attributes in the dataset into nominal attributes.

More

Capabilities

attributeIndices: 1

binRangePrecision: 6

bins: 3

debug: False

desiredWeightOfInstancesPerInterval: -1.0

doNotCheckCapabilities: False

findNumBins: False

ignoreClass: False

invertSelection: False

makeBinary: False

spreadAttributeWeight: False

useBinNumbers: False

useEqualFrequency: False

Open... Save... OK Cancel

Step 4 – Click on “Apply”.

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

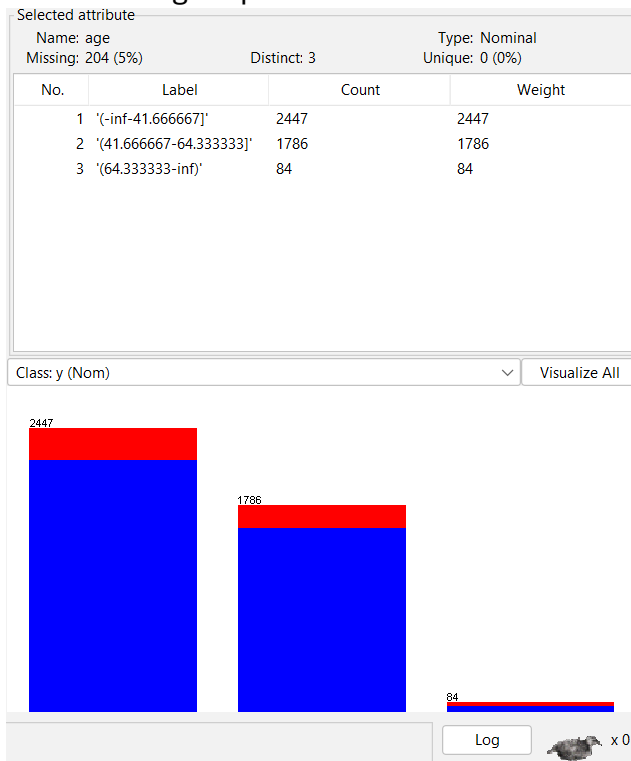
Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter

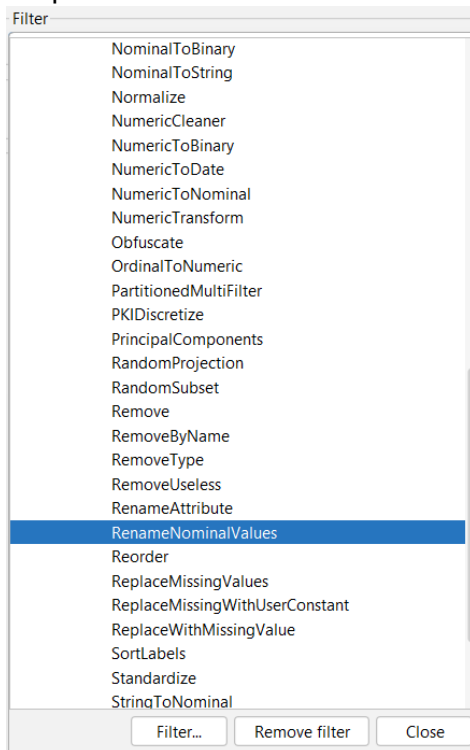
Choose Discretize -B 10 -M -1.0 -R first-last -precision 6 Apply Stop

Current relation		Selected attribute	
Relation: bank_dirty-weka.filters.unsupervised.attribute.Remove...	Attributes: 13	Name: age	Type: Numeric
Instances: 4521	Sum of weights: 4521	Missing: 204 (5%)	Unique: 4 (0%)
		Distinct: 67	

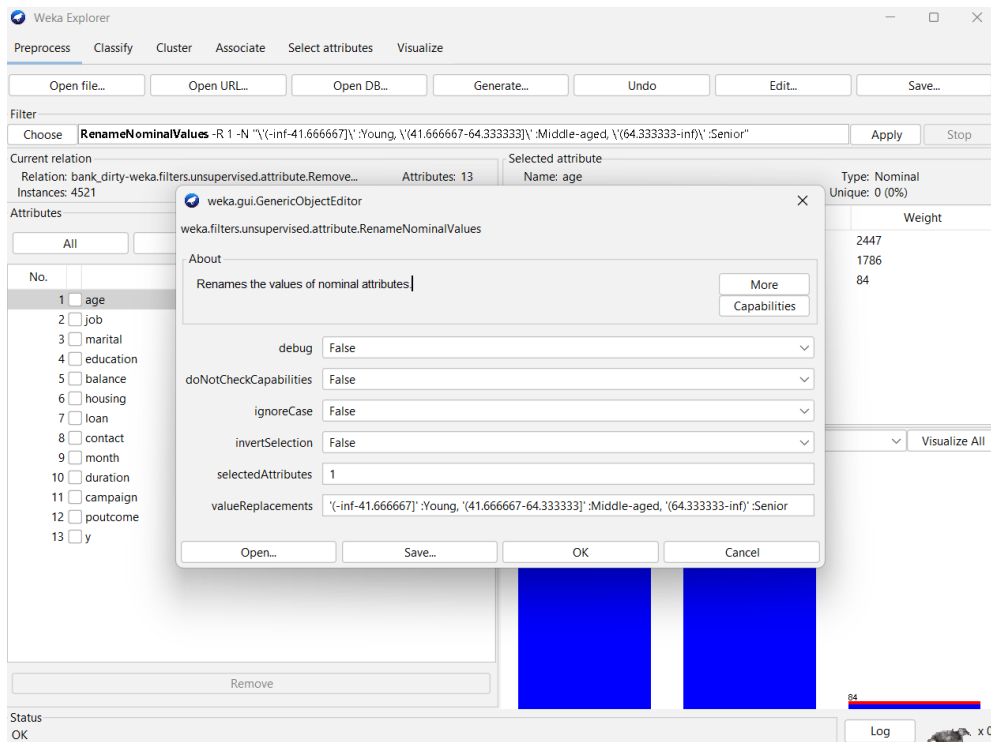
The following output will be shown.



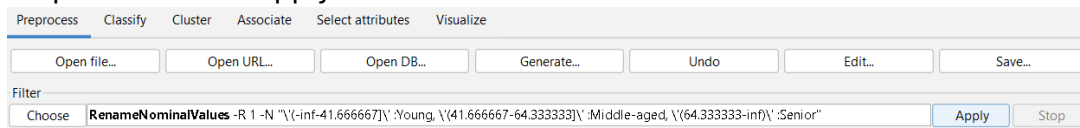
Step 5 – In order to rename the labels, select the “RenameNominalValues” filter.



Step 6 – Set the value of the selected attribute as 1, which is the index of the attribute “age” and enter the value replacements as shown in the screenshot to rename the labels.



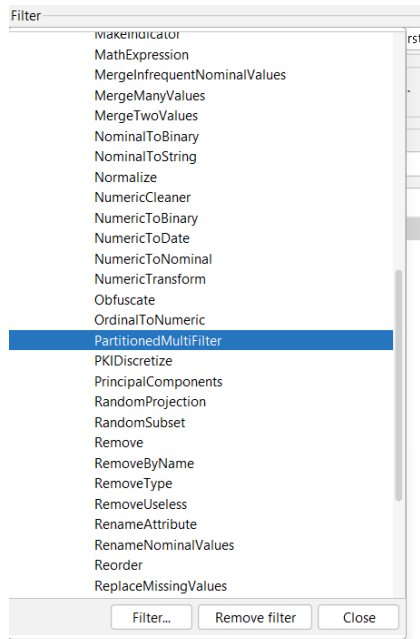
Step 7 – Click on “Apply”.



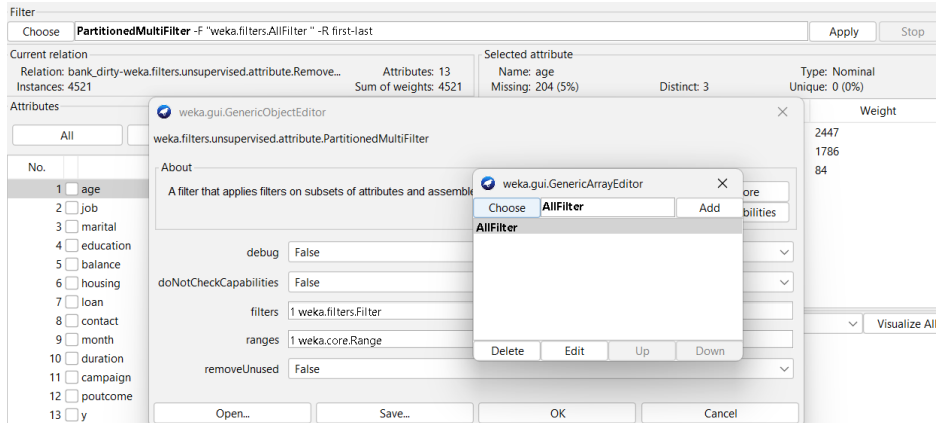
Verify that the labels have changed from the bin boundaries to Young, Middle-aged and Senior as shown below.



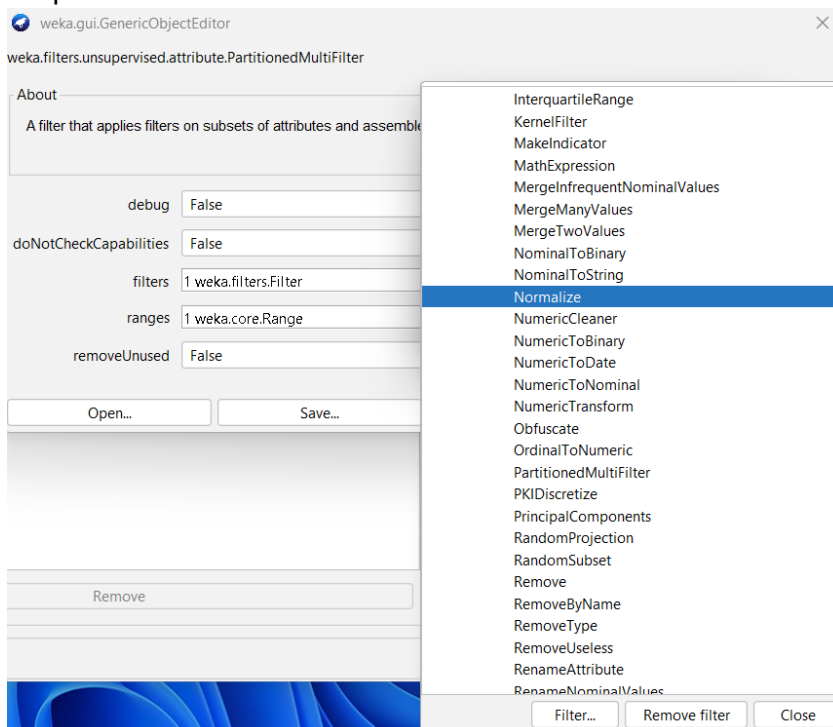
Step 8 – In order to normalize the attribute “balance” to have values between 0 and 1, select the “PartitionedMultiFilter” from the filter list.



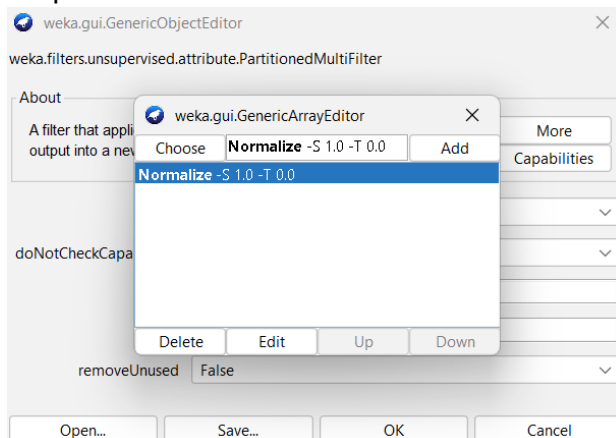
Step 9 – Click on the filter and then click on the “Choose” button.



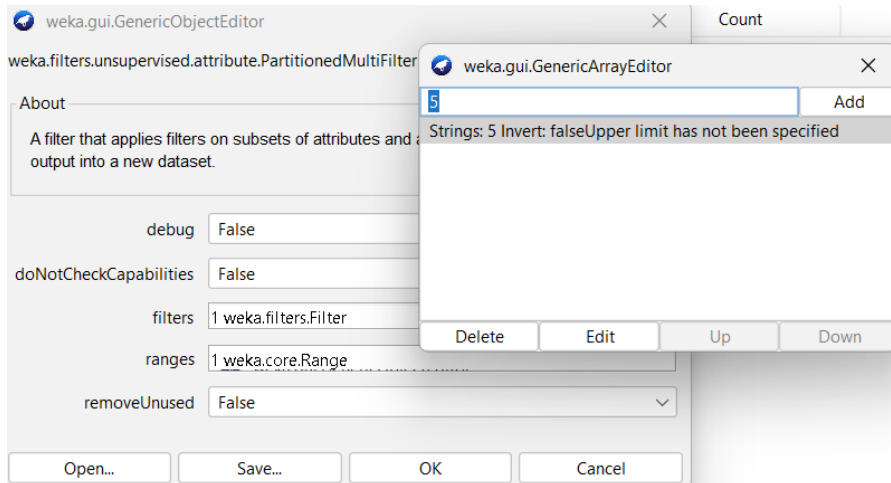
Step 10 – Select “Normalize” From the list shown below.



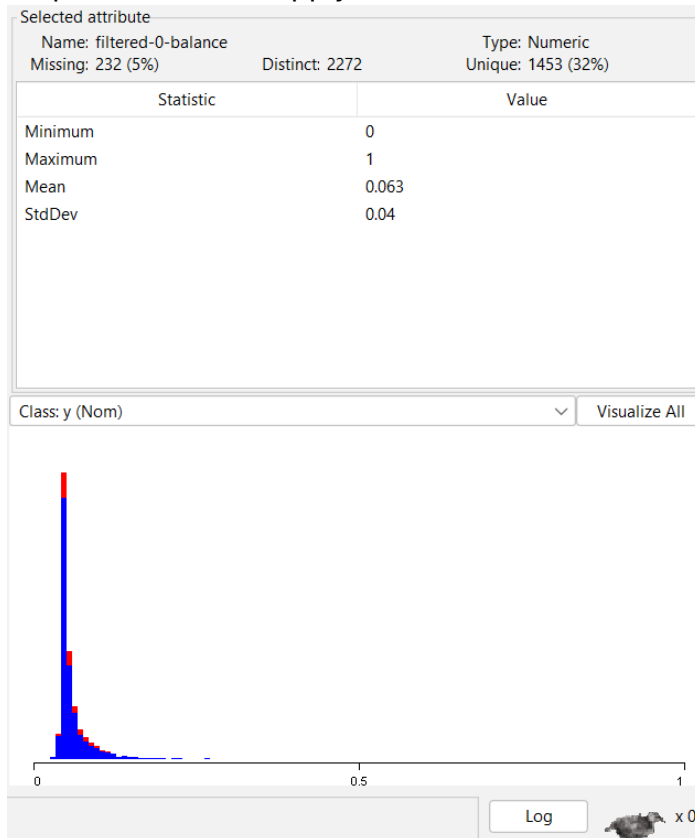
Step 11 – Click on the “Add” button to add the normalize filter.



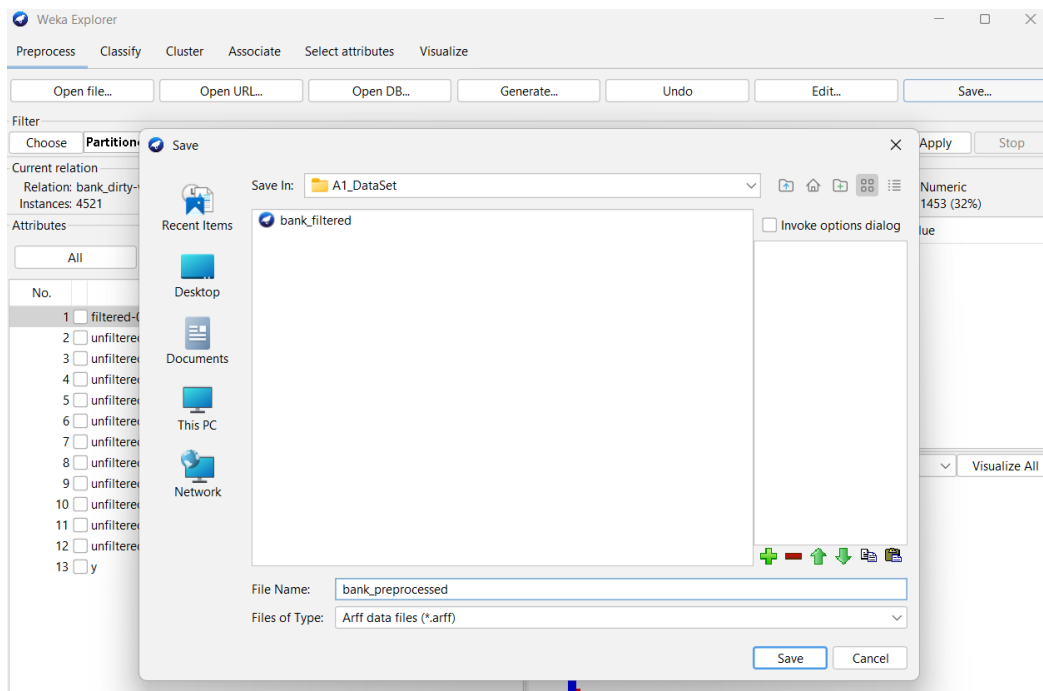
Step 12 – Click on “ranges” and then set the value as the index of the attribute “balance”. In this case the value is 5. Then click on “Add”.



Step 13 – Click on “Apply” to normalize the attribute “balance”.

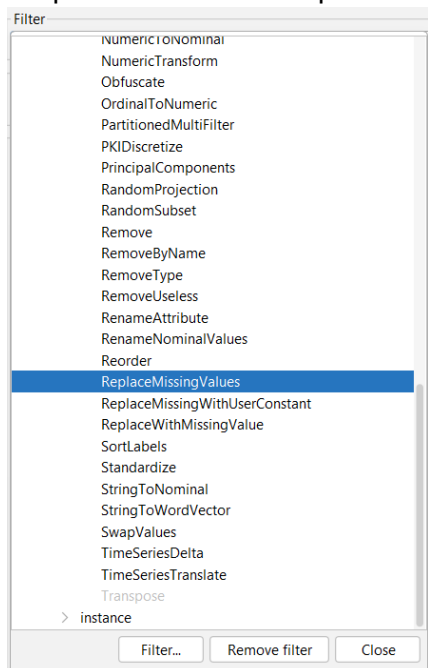


Step 13 – Click on “Save” button and rename the data file to “bank_preprocessed” and then click on “Save” button to save the file as shown below.

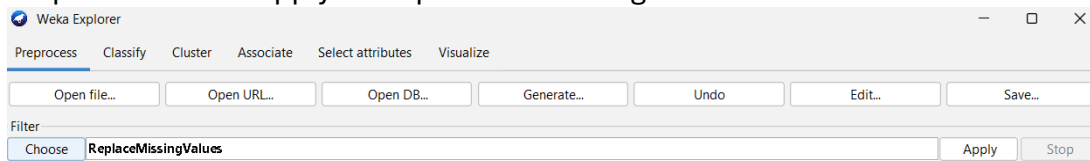


1.3 Handling Missing Values:

Step 1 – Select the “ReplaceMissingValues” filter from the filter list.



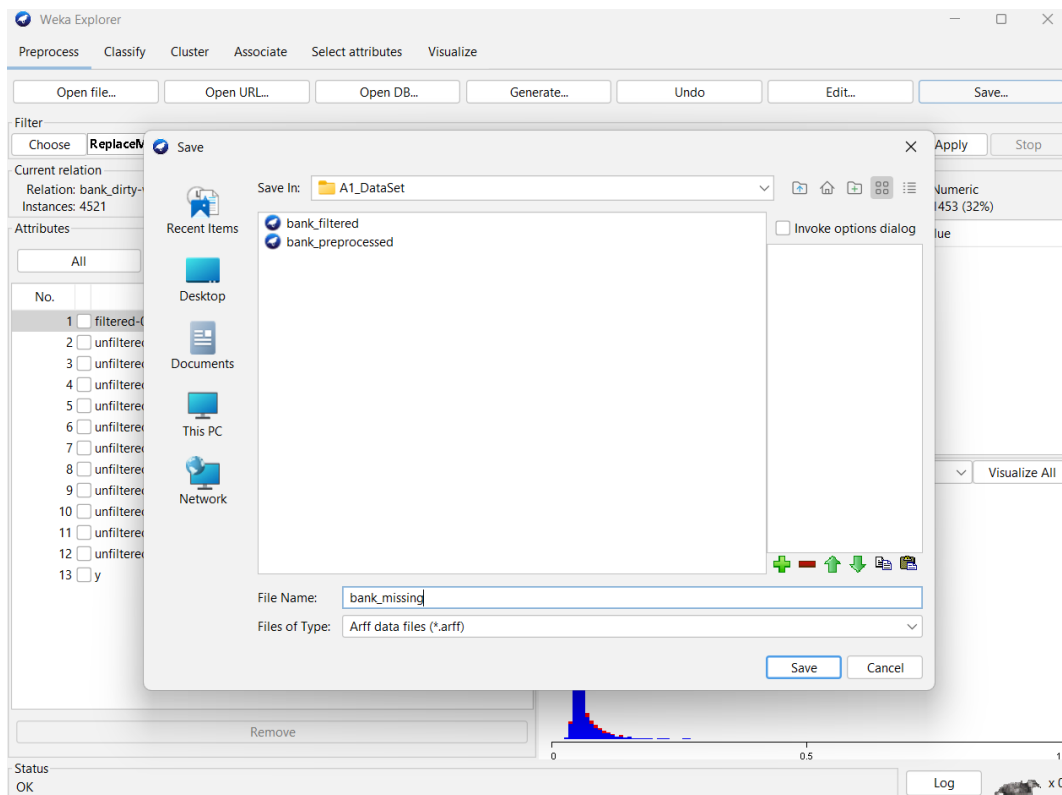
Step 2 – Click on “Apply” to replace all missing values in the dataset.



No.	1: filtered-0-balance	2: unfiltered-age	3: unfiltered-job	4: unfiltered-marital	5: unfiltered-education	6: unfiltered-housing	7: unfiltered-loan	8: unfiltered-filter	9: unfiltered-
	Numeric	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal
1	0.0684554569737319	Young	unemployed	married	primary	no	no	cellular	oct
2	0.1087502181178776	Young	services	married	secondary	yes	yes	cellular	may
3	.0625897638957866...	Young	management	single	tertiary	yes	no	cellular	apr
4	.0642810163621964...	Young	management	married		yes	yes	unknown	jun
5	.0444692017556811...	Middle-aged	blue-collar	married	secondary	yes	no	unknown	may
6	.0544959128065395...	Young	management	single	tertiary	no	no	cellular	feb
7	.0485899518127273...	Young	self-employed	married	tertiary	yes	no	cellular	may
8	.0464423296331592...	Young	technician	married	secondary	yes	no		may
9	.0474356048912095...	Young	entrepreneur	married	tertiary	yes	no	unknown	may
10	.0432880095569187	Middle-aged	services	married	primary	yes	yes	cellular	apr
11	.1702930162011248...	Young	services	married	secondary	yes	no	unknown	may
12	.0480127783519684...	Middle-aged	admin.	married	secondary	yes	no	cellular	apr
13	.0593549079878122...	Young	technician	married	tertiary	no	no	cellular	aug
14	.0512073663440759...	Young	student	single	secondary	no	no	cellular	apr
15	.0493013516597092...	Young	blue-collar	married	secondary	yes	yes	cellular	jan
16	.047073193648074	Young	management	married	tertiary	no	yes	cellular	aug
17	.09911396088643105...	Middle-aged	technician	married	secondary	no	no	cellular	aug
18	.0755694554435510...	Young	admin.	single	tertiary	yes	no	cellular	apr
19	.0415027986201527...	Young	blue-collar	single	primary	yes	no	unknown	may
20	.0462409900538247...	Young	services	married	secondary	no	no	cellular	jul
21		Young	management	divorced	unknown	yes	no	cellular	
22	.0446839639736739...	Middle-aged	management	divorced	tertiary	no	no		nov
23	.0458920014496449...	Middle-aged	services	single	secondary	no	no		jun

No.	1: filtered-0-balance	2: unfiltered-age	3: unfiltered-job	4: unfiltered-marital	5: unfiltered-education	6: unfiltered-housing	7: unfiltered-loan	8: unfiltered-contact	9: unfil
	Numeric	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	Nominal	N
1	0.06484554569737319	Young	unemployed	married	primary	no	no	cellular	oct
2	0.10875021811178776	Young	services	married	secondary	yes	yes	cellular	may
3	.0625897638957866...	Young	management	single	tertiary	yes	no	cellular	apr
4	0.0642810163621964...	Young	management	married	secondary	yes	yes	unknown	jun
5	0.0444692017556811...	Middle-aged	blue-collar	married	secondary	yes	no	unknown	may
6	0.0544959128065395...	Young	management	single	tertiary	no	no	cellular	feb
7	0.0485899518127273...	Young	self-employed	married	tertiary	yes	no	cellular	may
8	0.0464423296331592...	Young	technician	married	secondary	yes	no	cellular	may
9	0.0474356048912095...	Young	entrepreneur	married	tertiary	yes	no	unknown	may
10	0.0432880095569187	Middle-aged	services	married	primary	yes	yes	cellular	apr
11	.1702930162011248...	Young	services	married	secondary	yes	no	unknown	may
12	0.0480127783519684...	Middle-aged	admin.	married	secondary	yes	no	cellular	apr
13	0.0593549079878122...	Young	technician	married	tertiary	no	no	cellular	aug
14	0.0512073663440759...	Young	student	single	secondary	no	no	cellular	apr
15	0.0493013516597092...	Young	blue-collar	married	secondary	yes	yes	cellular	jan
16	0.0470731936484074	Young	management	married	tertiary	no	yes	cellular	aug
17	0.0991396088643105...	Middle-aged	technician	married	secondary	no	no	cellular	aug
18	0.0755694554435510...	Young	admin.	single	tertiary	yes	no	cellular	apr
19	0.0415027986201527...	Young	blue-collar	single	primary	yes	no	unknown	may
20	0.0462409900538247...	Young	services	married	secondary	no	no	cellular	jul
21	0.063355414486652	Young	management	divorced	unknown	yes	no	cellular	nov
22	0.0446839639736379...	Middle-aged	management	divorced	tertiary	no	no	cellular	nov
23	0.0458920014496449...	Middle-aged	services	single	secondary	no	no	cellular	jun

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Summary of how missing values were handled:

The missing values in the provided dataset were handled via the “ReplaceMissingValues” filter in the WEKA application. This filter handles two main types of attributes, Categorical (nominal) and numerical. In terms of handling missing values of a numerical attribute, the filter computes the mean (average) of that particular attribute and replaces the missing values with the mean value. On the other hand, the filter handles missing values of a nominal attribute by computing the mode of the attribute or the class mode and replaces all the missing values with the computed mode value.

2.0 Classification Using J48, PART, Logistic Regression, and Random Forest

The "bank_preprocessed.arff" dataset generated during Question 1 was loaded into WEKA for classification. The dataset contained 4,521 instances and 13 attributes, with y selected as the target class. The four classifiers evaluated were J48, PART, Logistic Regression, and Random Forest. All classifiers were evaluated using 10-fold cross-validation to provide a consistent basis for comparing their performance. The preprocessed dataset used for these experiments is shown in Figure 1.

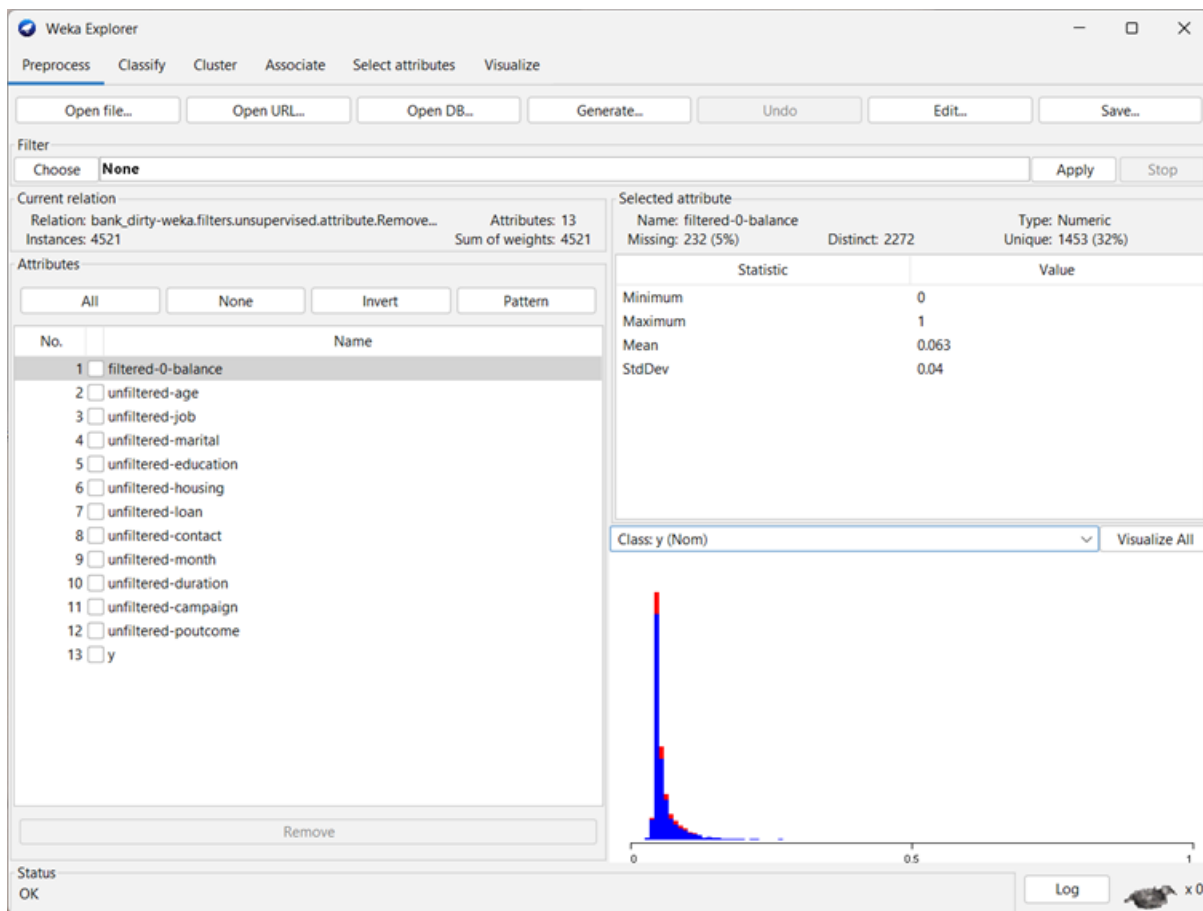


Figure 1 Preprocessed Bank Marketing Dataset Loaded in WEKA

2.1 J48 Decision Tree:

The J48 classifier was applied to bank_preprocessed.arff with y as the target attribute. As shown in Figure 2, 10-fold cross-validation was selected before the classifier was executed. The trained J48 model was then saved for later use.

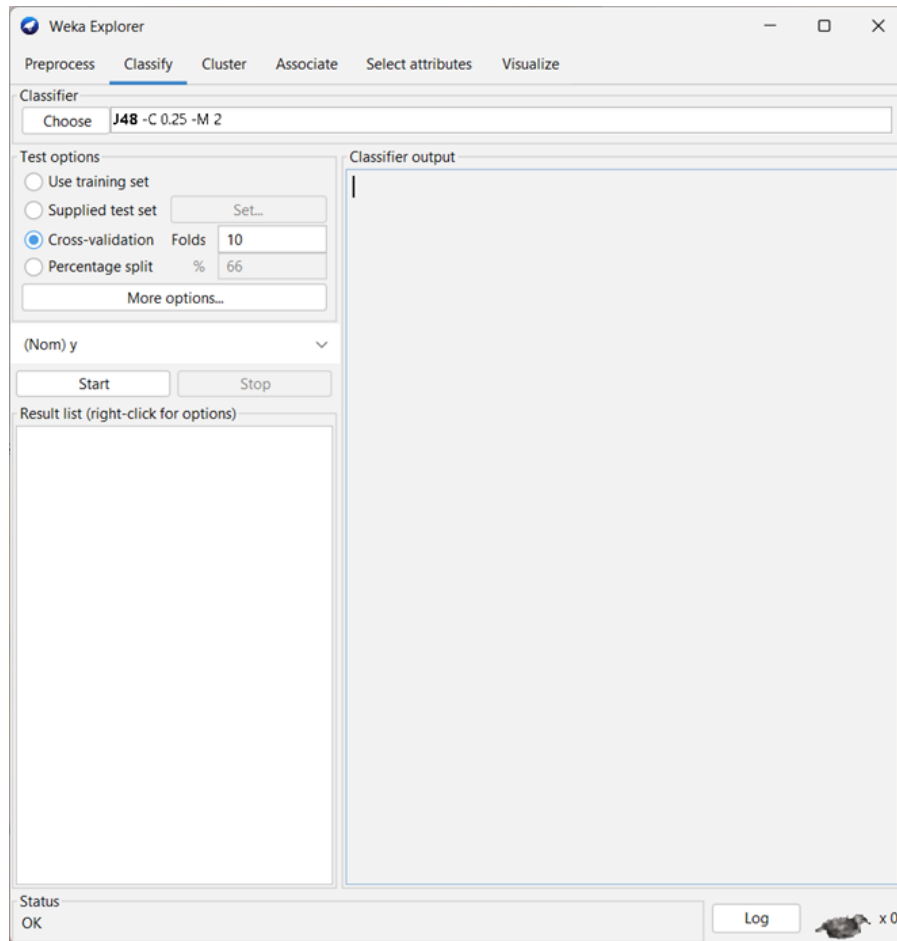


Figure 2 J48 Configuration Using 10-Fold Cross-Validation

The J48 classifier correctly classified 4,043 out of 4,521 instances, producing an overall accuracy of 89.4271%. The generated decision tree contained 43 leaves and had a total tree size of 57.

The results also show a substantial difference in performance between the two target classes. For y = no, J48 achieved a recall of 0.971, precision of 0.915, and F1-score of

0.942. In comparison, for y = yes, recall was only 0.307, precision was 0.578, and the F1-score was 0.401.

The confusion matrix further demonstrates this difference:

Actual Class	Predicted (Yes)	Predicted (No)
Yes	160	361
No	117	3883

Of the 4,000 clients who did not subscribe, J48 correctly classified 3,883. However, of the 521 clients who subscribed, only 160 were correctly classified, while 361 were incorrectly predicted as no. Therefore, although the overall accuracy is high, J48 is considerably better at identifying non-subscribers than subscribers.

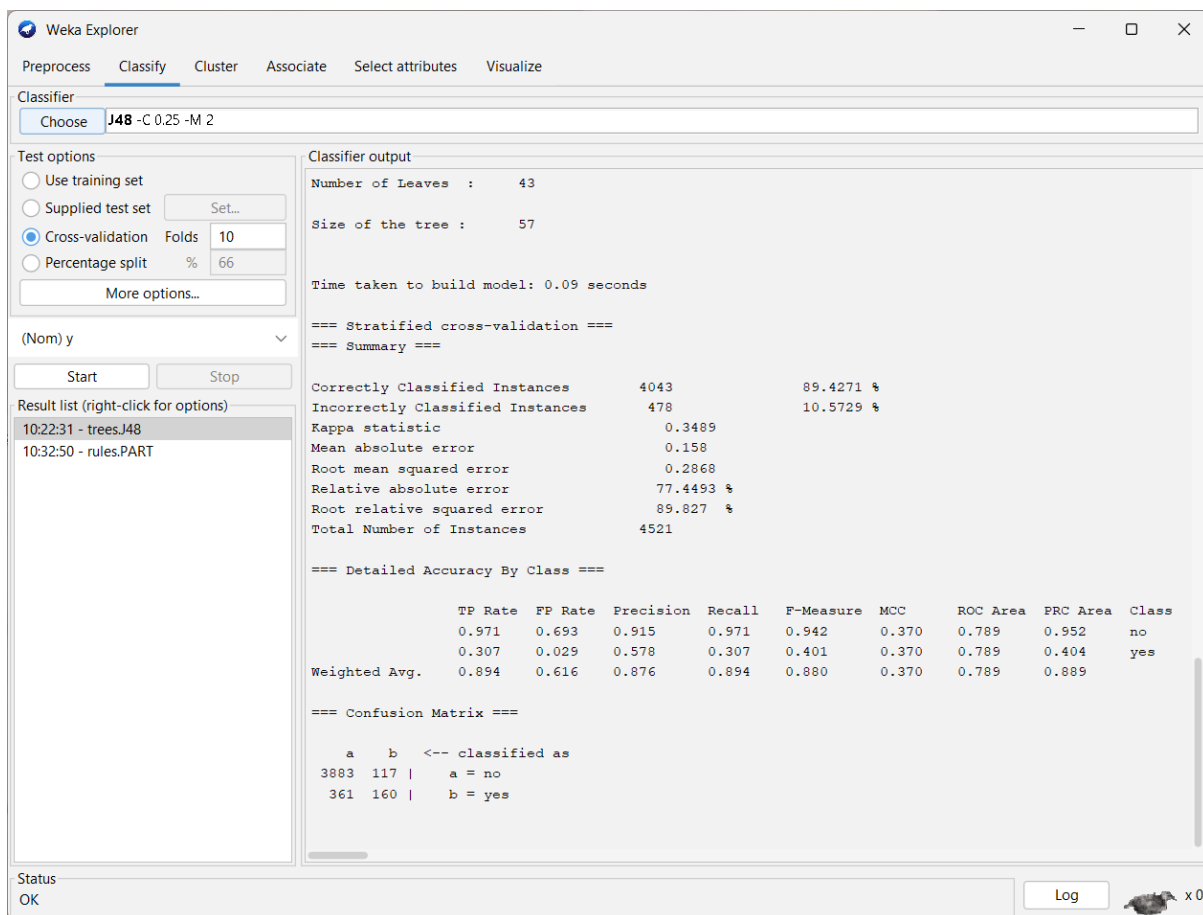


Figure 1 J48 Classification Results

The generated J48 tree is shown in Figure 4. The first split is based on duration, indicating that contact duration plays an important role in the classifier's decision structure. Additional branches use attributes including poutcome, marital, job, balance, campaign and contact-related information to further separate clients into the yes and no classes.



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the major decision factors, but its full decision structure becomes less straightforward as the tree becomes deeper.

2.2 PART Rule Analysis:

The PART classifier was applied to the same dataset using 10-fold cross-validation and y as the target attribute. The trained PART model was then saved for later use. PART generated a total of 125 classification rules. Due to the length of the rule set, the complete PART rules are provided in the Appendix.

PART correctly classified 3,994 of the 4,521 instances, giving an overall accuracy of 88.3433%.

For y = no, PART achieved:

- Precision = 0.918
- Recall = 0.954
- F1-score = 0.935

For y = yes, PART achieved:

- Precision = 0.492
- Recall = 0.344
- F1-score = 0.405

Its confusion matrix was:

Actual Class	Predicted (Yes)	Predicted (No)
Yes	179	342
No	185	3815

PART therefore correctly identified 179 of the 521 actual subscribers, compared with 160 for J48.

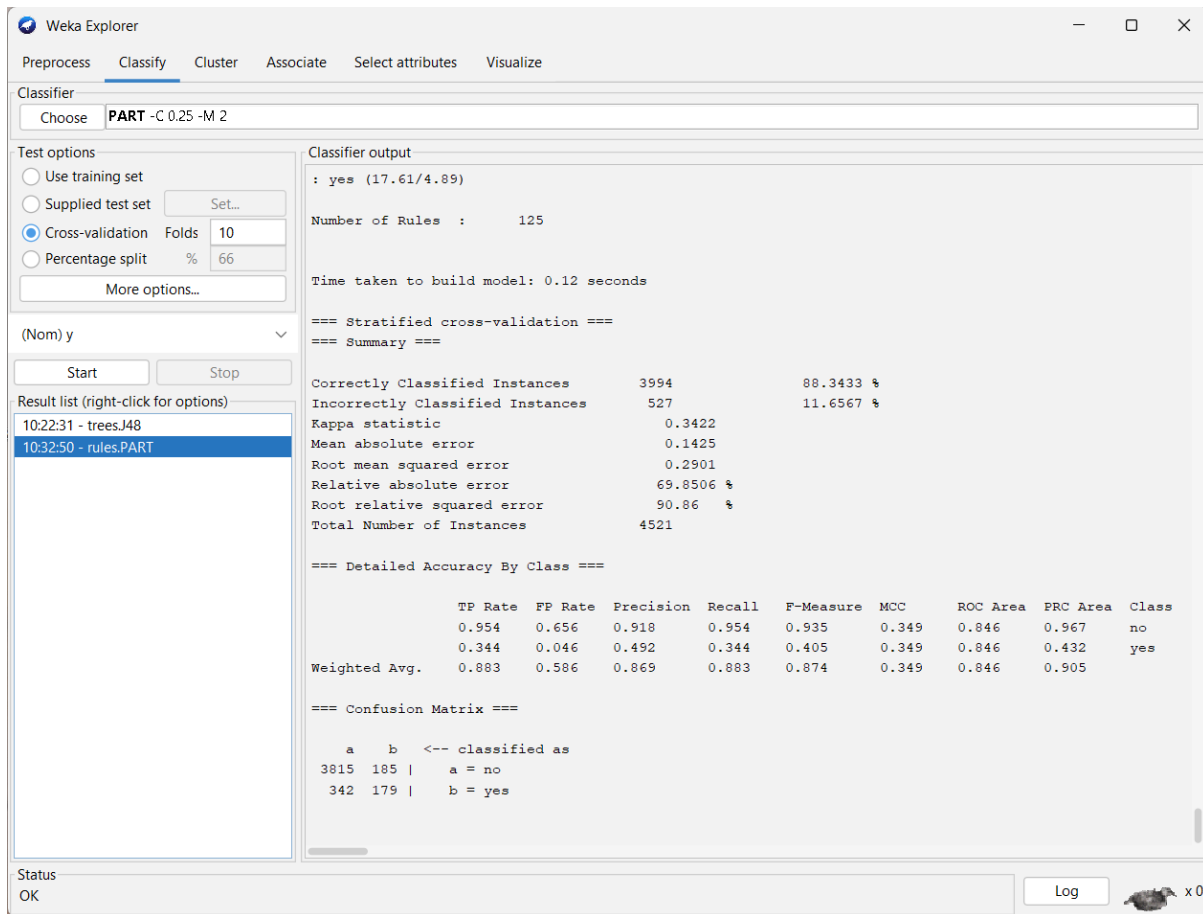


Figure 2 PART Rules and Classification Results

Comparison of PART and J48

J48 achieved a higher overall accuracy of 89.4271%, compared with 88.3433% for PART. J48 was also stronger for the majority no class, obtaining an F1-score of 0.942 compared with 0.935 for PART.

However, PART performed slightly better at identifying the minority yes class. Its recall for subscribers was 0.344, compared with 0.307 for J48, meaning PART correctly detected a greater proportion of customers who actually subscribed. PART also achieved a slightly higher yes F1-score of 0.405, compared with 0.401 for J48. The trade-off was lower yes precision: 0.492 for PART compared with 0.578 for J48.

In terms of simplicity, J48 produced 43 leaves with a tree size of 57, whereas PART generated 125 rules. PART rules are individually easy to understand because they are expressed as direct conditions leading to a classification. However, having 125 rules makes the complete rule set relatively lengthy. J48 provides a clearer overview of the major decision structure, but its lower branches become visually difficult to follow. Therefore,

neither model is completely simple: J48 is more compact overall, while individual PART rules can be easier to interpret in isolation.

2.3 Logistic Regression:

Logistic Regression was then applied using the same 10-fold cross-validation procedure. The classifier correctly classified 4,079 out of 4,521 instances, resulting in an overall accuracy of 90.2234%. This was higher than both J48 and PART.

For $y = \text{no}$, Logistic Regression obtained:

- Precision = 0.917
- Recall = 0.978
- F1-score = 0.947

For $y = \text{yes}$, it obtained:

- Precision = 0.654
- Recall = 0.322
- F1-score = 0.432

Its confusion matrix was:

Actual Class	Predicted (Yes)	Predicted (No)
Yes	168	353
No	89	3911

Therefore, Logistic Regression correctly identified 168 subscribers and incorrectly classified 353 subscribers as non-subscribers.

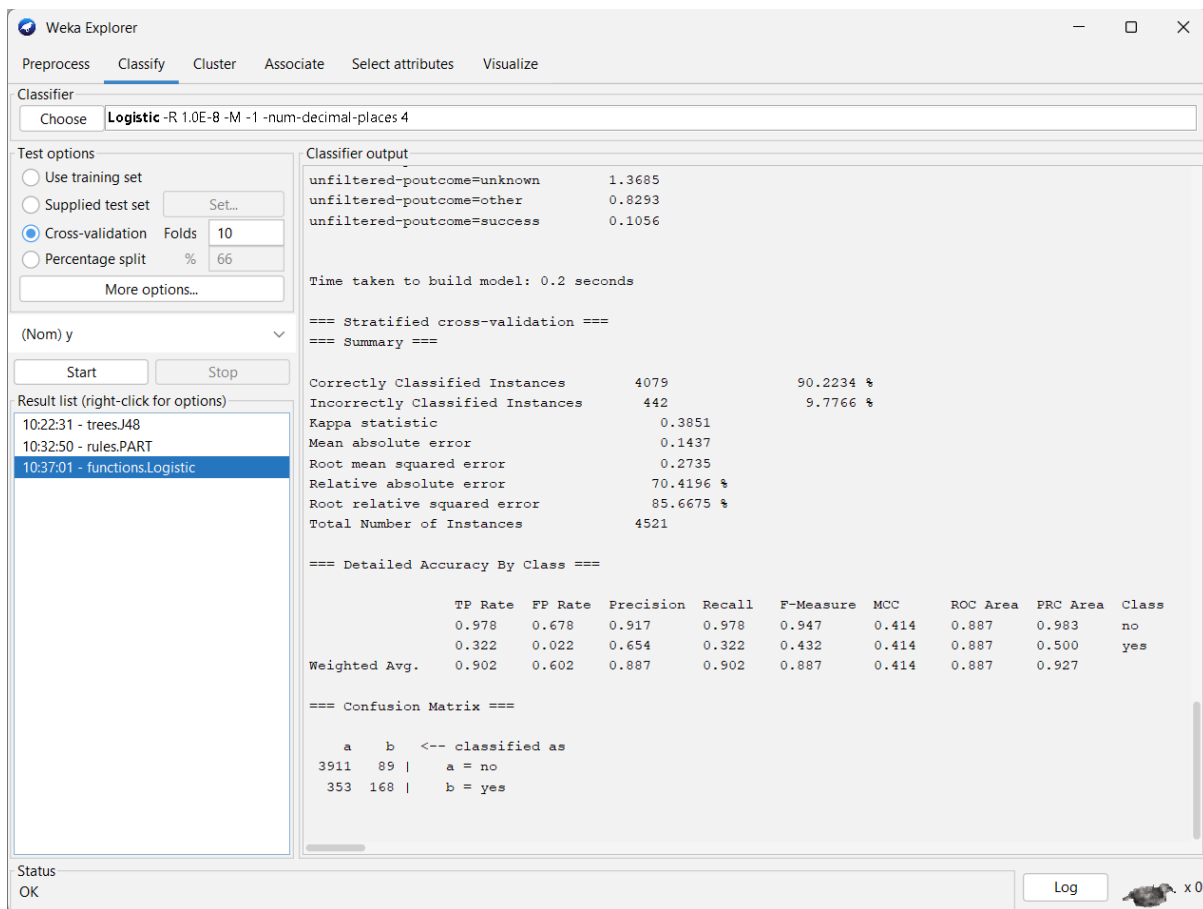


Figure 3 Logistic Regression Classification Results

Compared with J48 and PART, Logistic Regression achieved the highest overall accuracy. It also achieved the highest precision for yes, at 0.654, meaning its positive subscription predictions were more reliable than those made by J48 or PART.

PART nevertheless had a slightly better recall for yes (0.344) than Logistic Regression (0.322), meaning PART identified a slightly larger proportion of all actual subscribers.

Logistic Regression Coefficient Analysis

Figures 7 and 8 show the coefficients produced by Logistic Regression.

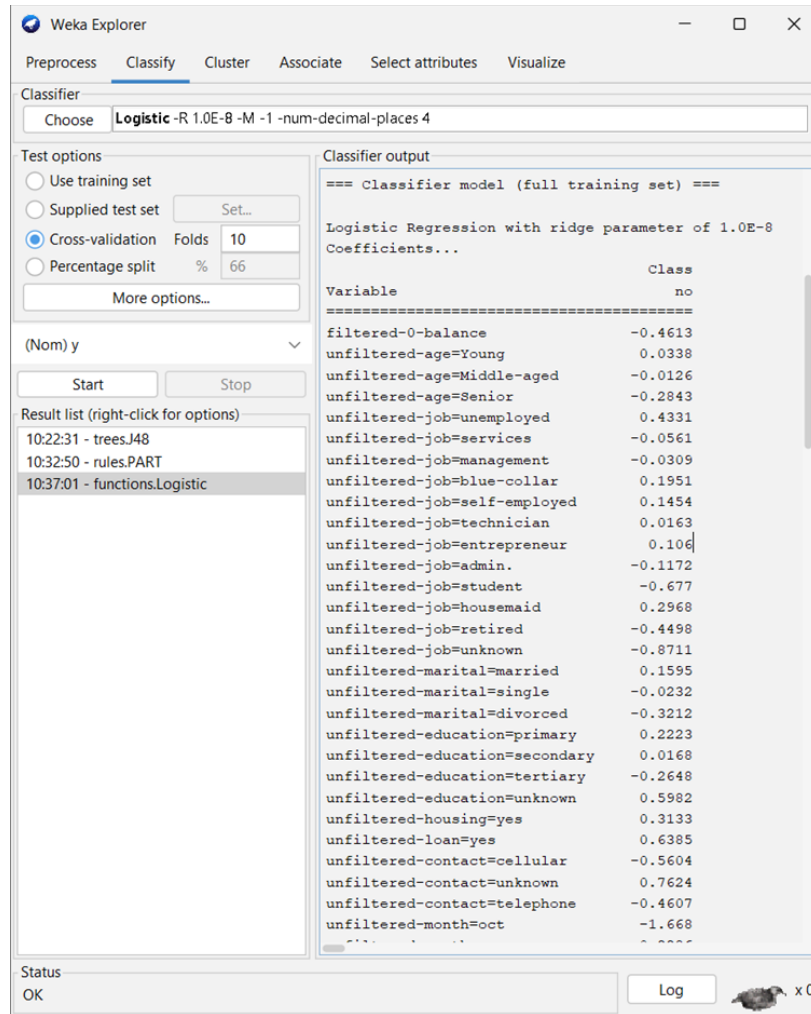


Figure 7 Logistic Regression Coefficient Summary part 1

unfiltered-month=may	0.2286
unfiltered-month=apr	-0.3433
unfiltered-month=jun	-0.6465
unfiltered-month=feb	-0.3638
unfiltered-month=aug	0.0415
unfiltered-month=jan	0.5773
unfiltered-month=jul	0.3959
unfiltered-month=nov	0.527
unfiltered-month=sep	-0.9613
unfiltered-month=mar	-1.7864
unfiltered-month=dec	-0.4187
unfiltered-duration	-0.0042
unfiltered-campaign	0.0618
unfiltered-poutcome=failure	0.2323
unfiltered-poutcome=unknown	0.3137
unfiltered-poutcome=other	-0.1871
unfiltered-poutcome=success	-2.2482
Intercept	3.3887

Figure 8 Logistic Regression Coefficient Summary part 2

An important point when interpreting these results is that WEKA displays the coefficients for class no in the coefficient output. Therefore, positive coefficients indicate movement toward the no class, while negative coefficients indicate movement away from no and consequently toward the yes subscription outcome.

Several notable coefficients can be observed.

poutcome = success

poutcome=success has a coefficient of -2.2482 , one of the strongest negative coefficients in the model. Since the coefficients are displayed for class no, this indicates a strong relationship in the direction of the yes class. Customers whose previous campaign outcome was successful therefore appear more likely to be associated with subscription. This also supports the decision made during preprocessing to retain poutcome.

Month

Several month categories have substantial coefficients. For example:

- March = -1.7864
- October = -1.6680
- September = -0.9613

These negative coefficients indicate movement away from the no class and toward yes. In contrast, January has a coefficient of 0.5773 and November 0.5270 , indicating movement toward the no class.

Contact

contact=cellular has a coefficient of approximately -0.5604 , while contact=unknown has a positive coefficient of approximately 0.7624 . This suggests that contact type contributes differently to the predicted subscription outcome.

Job

Some job categories also show notable effects. For example, job=student has a coefficient of approximately -0.6770 , while job=unemployed has a positive coefficient of approximately 0.4331 .

Duration

Duration has a coefficient of approximately -0.0042 . Although this appears small, duration is measured numerically, so the effect accumulates as contact duration increases. Its negative direction indicates movement away from no and toward subscription as duration increases.

This finding is also consistent with the J48 tree, where duration formed the root split.

Categorical and Numerical Attributes

Logistic Regression successfully handled the mixture of categorical and numerical attributes in the dataset but represents them differently from J48 and PART.

Numerical attributes such as balance, duration and campaign contribute numerically to the Logistic model. Categorical attributes such as job, marital status, education, contact, month and outcome are represented through separate category coefficients. This allows the model to estimate how individual categories contribute to the classification.

J48 and PART, in contrast, represent these relationships directly as tree splits or IF-THEN rules. This generally makes individual decisions easier for a human to follow. Logistic Regression provides a more compact mathematical representation, but interpreting a large number of categorical coefficients can be less intuitive.

In this dataset, Logistic Regression handled the combination effectively in terms of predictive performance, achieving 90.2234% accuracy, which was higher than J48 and PART. However, the tree and rule models provide more direct explanations of individual classification decisions.

2.4 Analysis of Misclassification:

WEKA's Visualize classifier errors function was used to investigate the errors produced by J48, PART, and Logistic Regression. Four attribute relationships were examined consistently across all three classifiers:

- Duration versus Balance
- Age versus Duration
- Job versus Duration
- Contact versus Duration

This allows similar error patterns to be compared across the three models.

A major issue affecting all three classifiers is the imbalance of the target variable. There are 4,000 no instances compared with only 521 yes instances. Consequently, the classifiers have considerably more examples from which to learn the no class.

This is reflected in their recall:

Model	Recall (No)	Recall (Yes)
J48	0.971	0.307
PART	0.954	0.344

Logistic Regression	0.978	0.322
----------------------------	--------------	--------------

All three classifiers therefore identify the no class very successfully but struggle considerably more with customers who actually subscribe.

J48 Misclassification Analysis

Figures 9–12 show the J48 error visualizations.

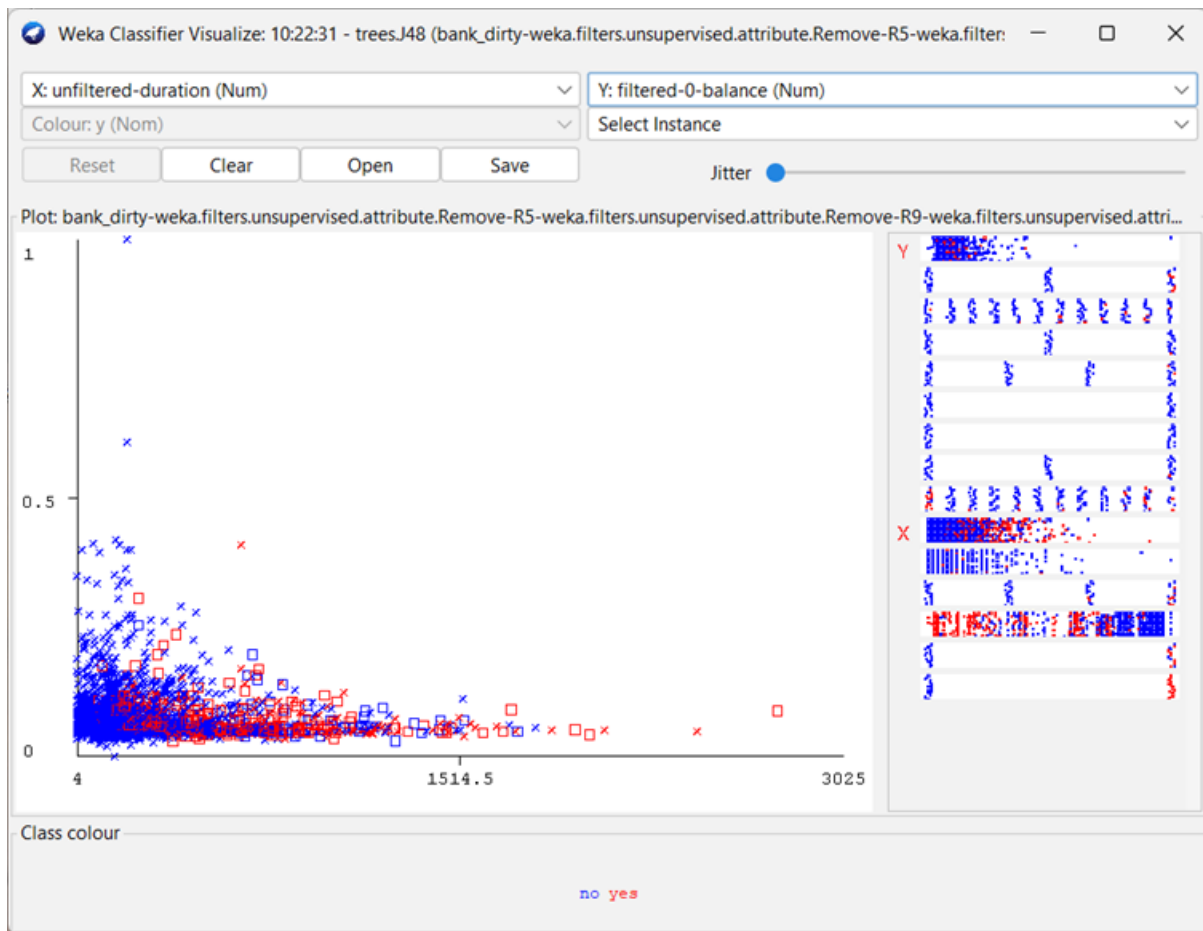


Figure 9 Duration vs Balance

The Duration versus Balance plot shows that most observations are concentrated toward the lower balance range, with many observations also concentrated at shorter contact durations. Misclassified instances appear within this dense region, indicating that customers with similar balance and duration characteristics can belong to different subscription outcomes.

The visualization also suggests that duration provides stronger separation as its value increases, while the densely populated lower-duration region contains greater overlap

between the two classes. This is consistent with duration appearing as the root attribute in the J48 tree.

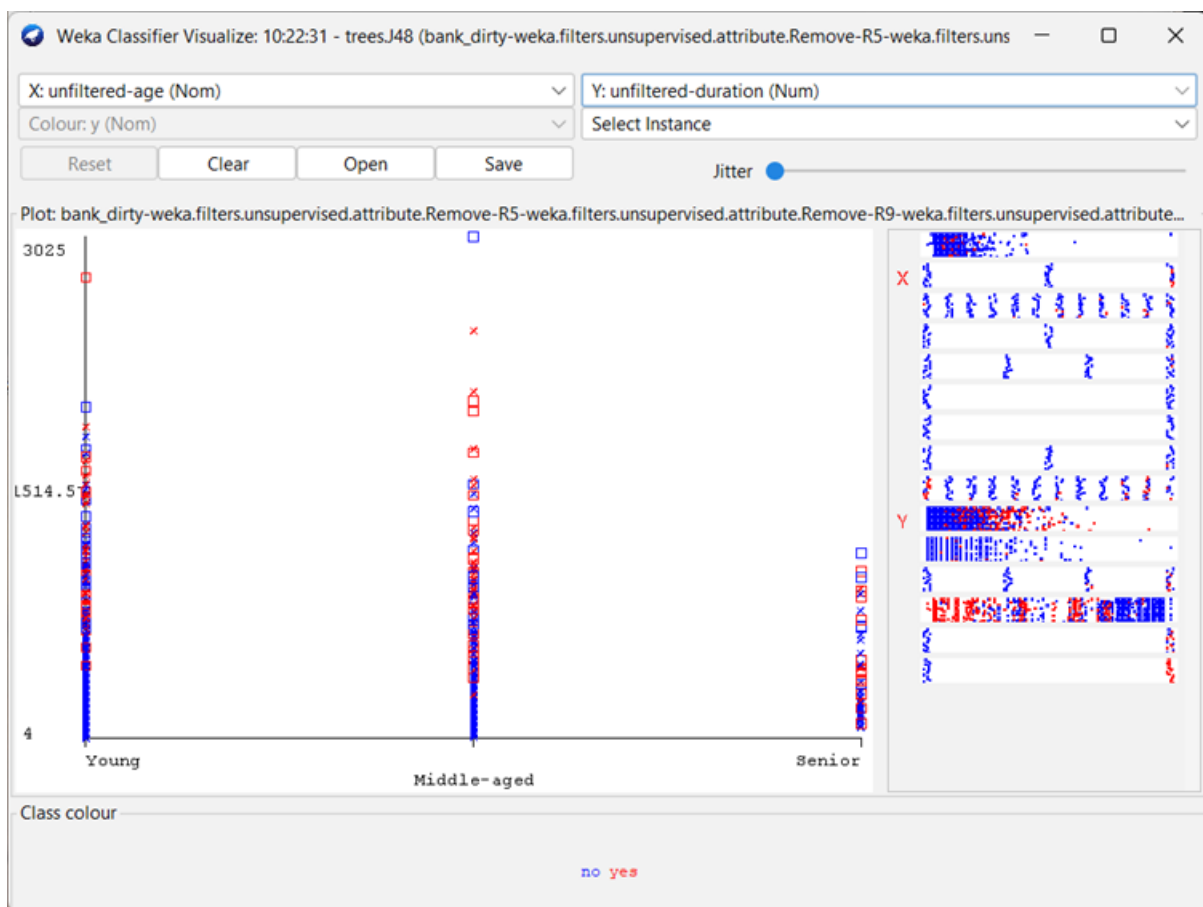
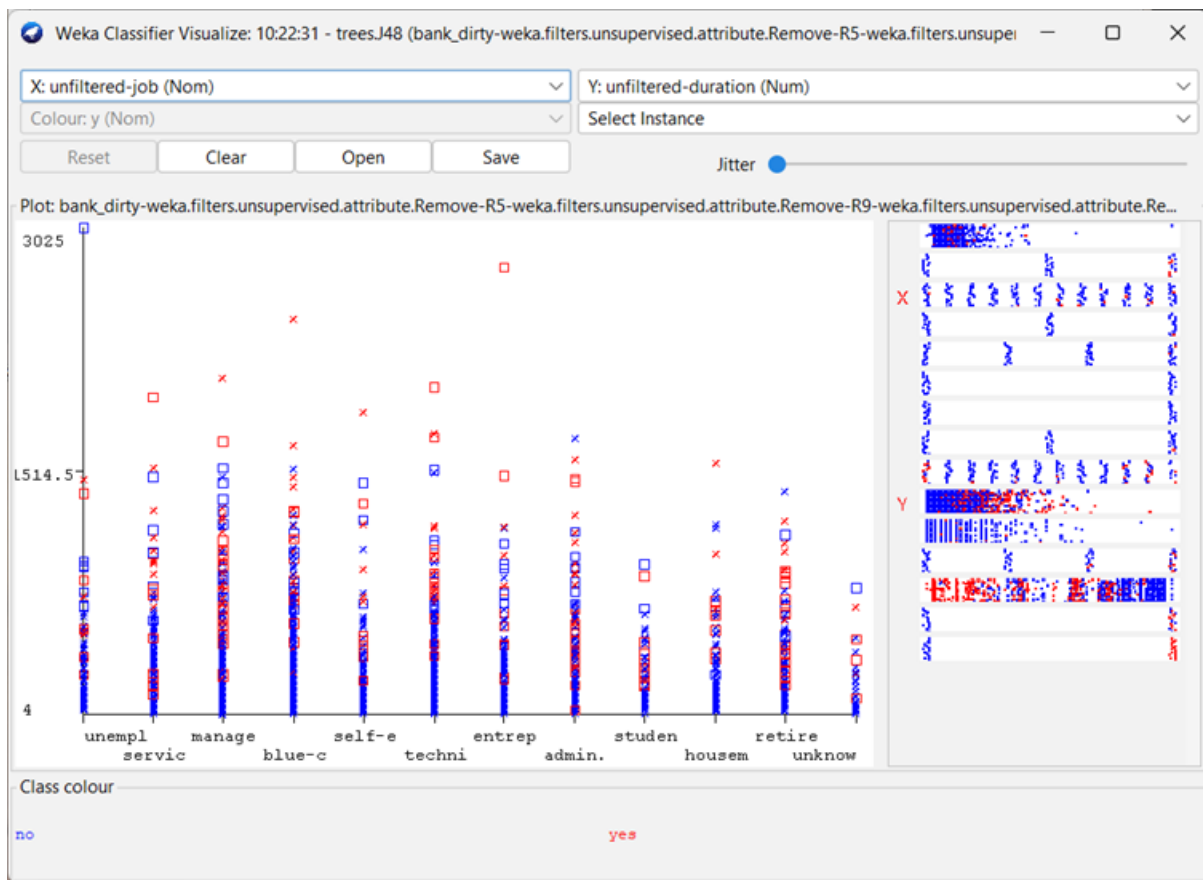


Figure 10 Age vs Duration

For Age versus Duration, misclassifications occur across all three age categories: Young, Middle-aged and Senior. Young and Middle-aged clients contain substantially more observations and consequently also visibly contain more errors. The plot does not show age alone providing a clear separation between subscribers and non-subscribers.



For Job versus Duration, errors can be observed across multiple job categories rather than being restricted to one occupation. This suggests that job by itself is insufficient to clearly determine whether a client will subscribe. Duration appears to provide additional differentiation within the job groups.

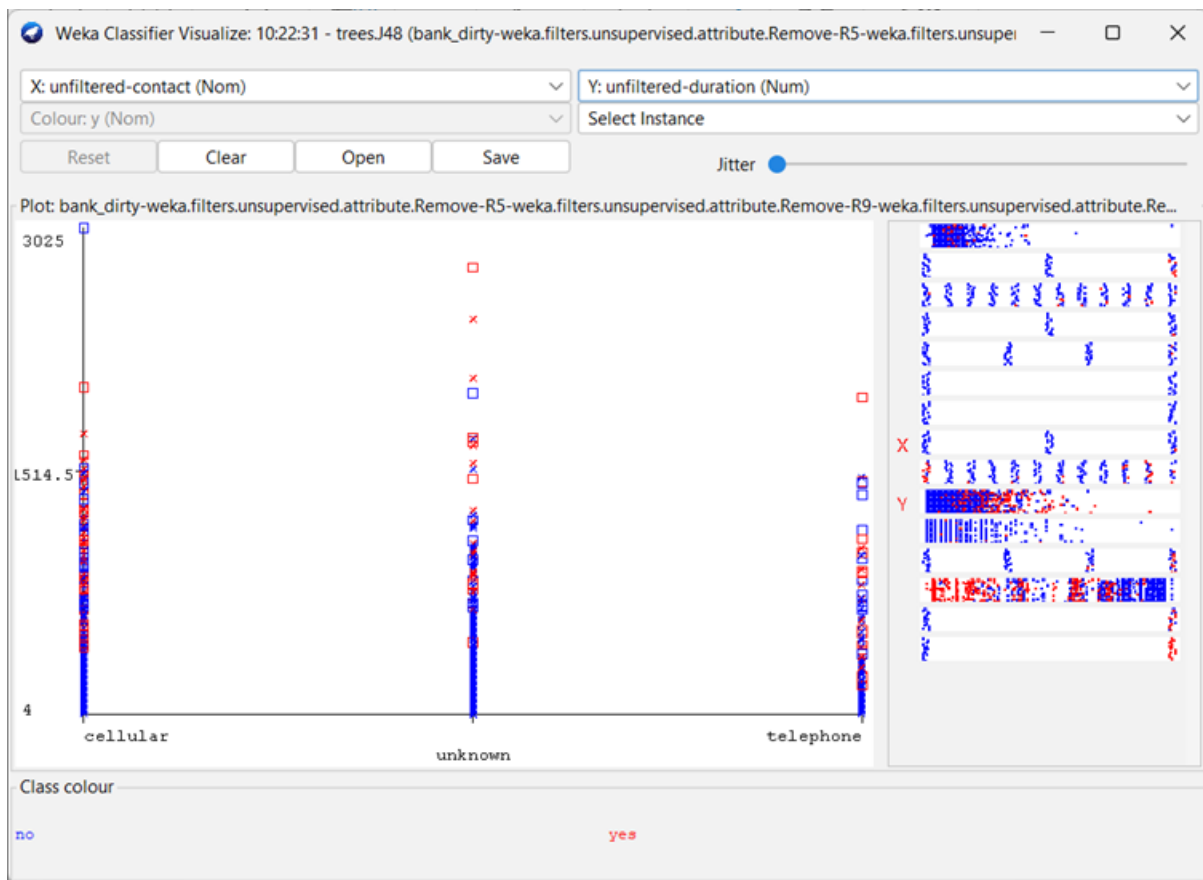


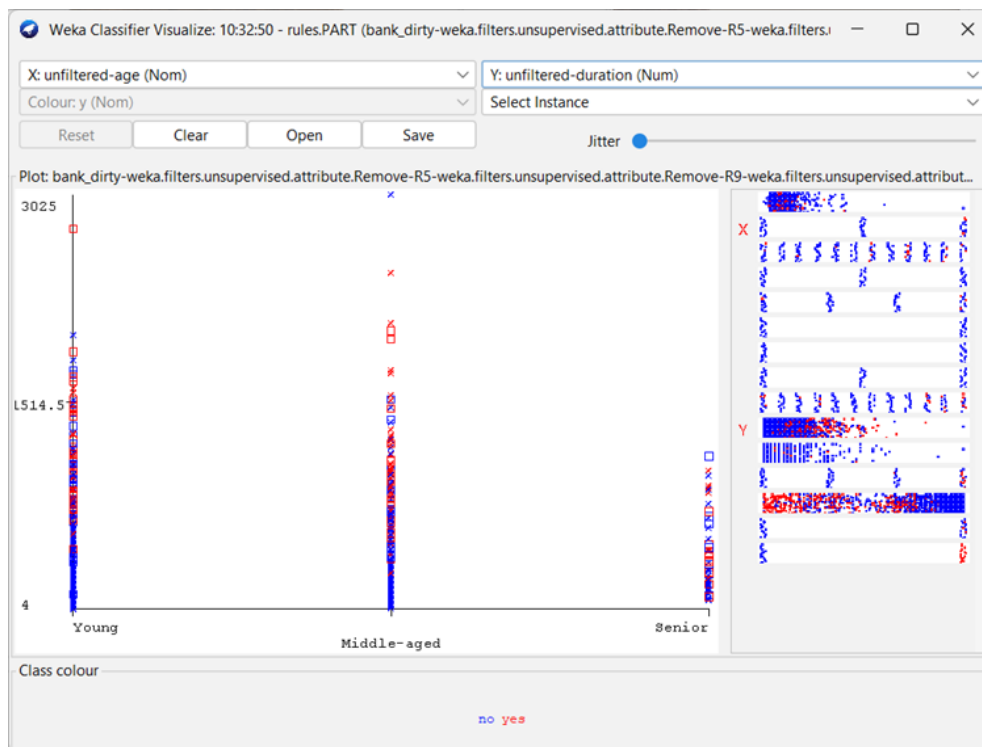
Figure 12 Contact vs Duration

Similarly, the Contact versus Duration plot contains classification errors across cellular, unknown, and telephone contacts. Cellular contains the largest concentration of observations and errors, but this should not automatically be interpreted as cellular contact causing errors because it also contains many more observations.

Overall, the J48 visualizations indicate substantial overlap between subscriber and non-subscriber characteristics, particularly within densely populated regions of the dataset.

PART Misclassification Analysis

Figures 13–16 show the corresponding PART error visualizations:



The age visualization again shows errors across all age categories rather than a single clearly problematic group. Likewise, the job and contact plots show that misclassifications occur across several categories.

Logistic Regression Misclassification Analysis

[illegible]

37

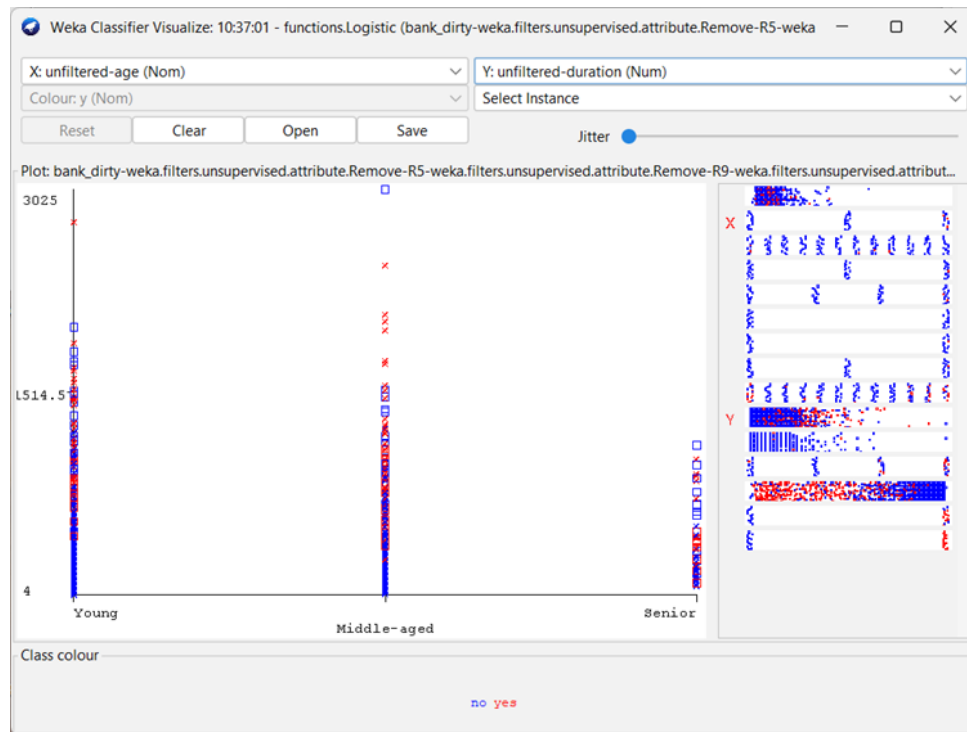


Figure 18 Age vs Duration

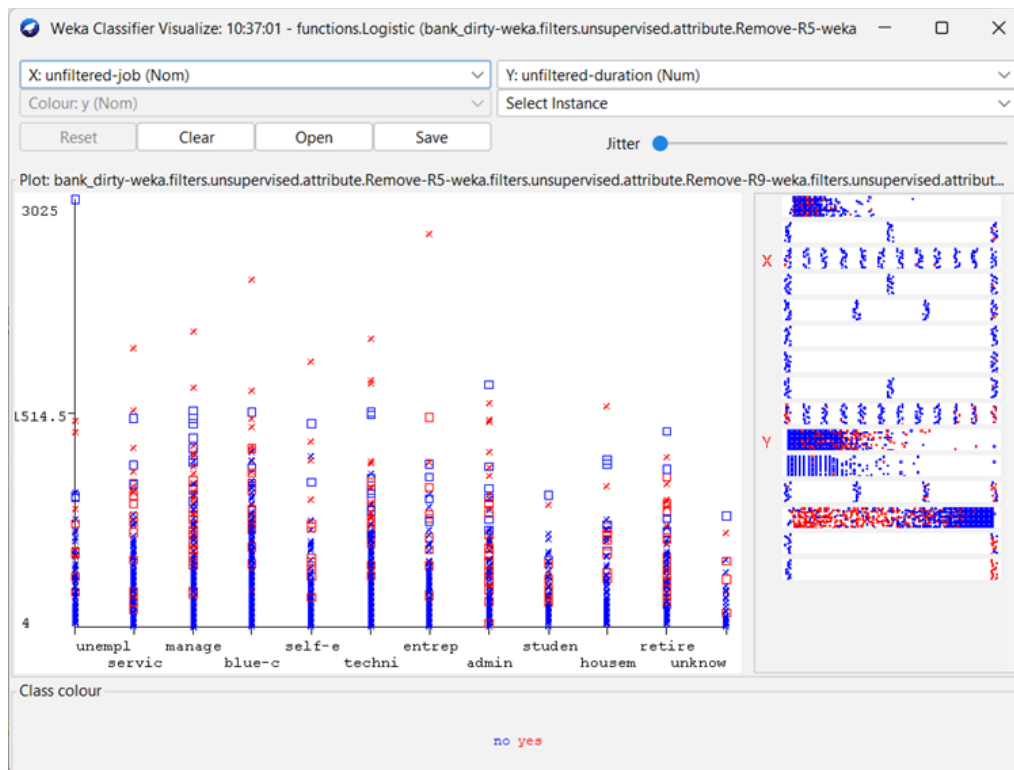


Figure 19 Job vs Duration

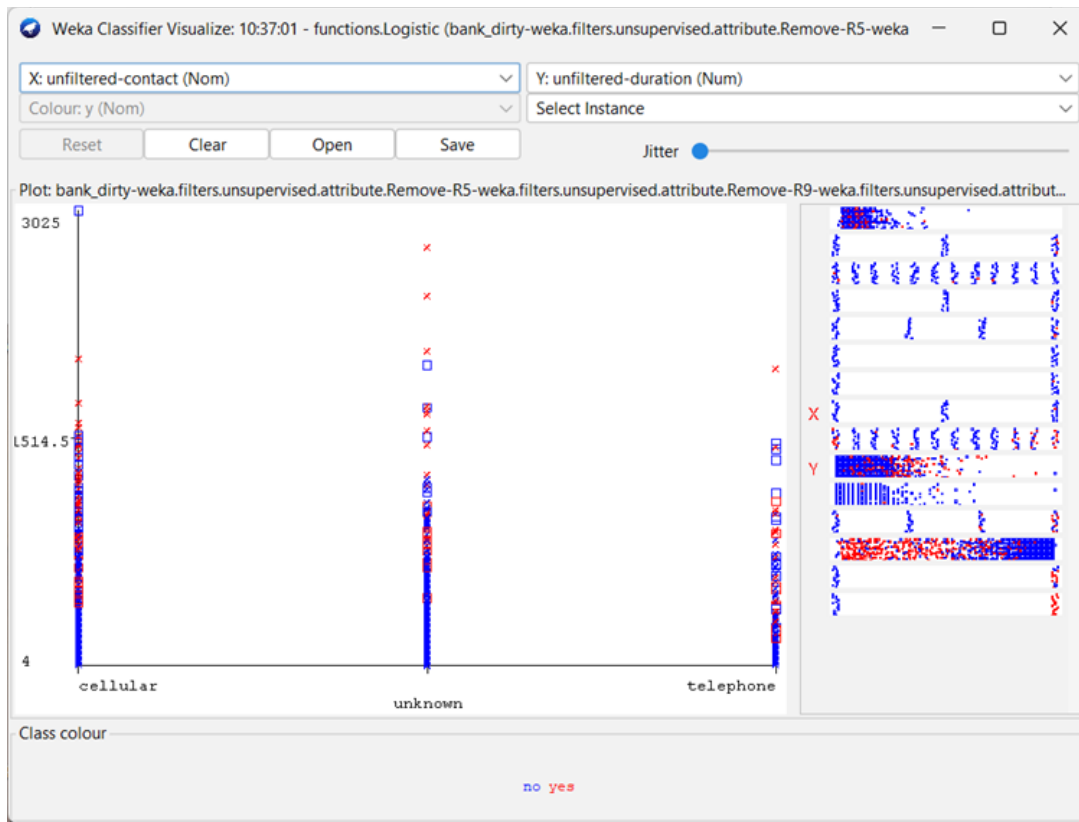


Figure 20 Contact vs Duration

The Logistic Regression plots again show considerable overlap between the two target classes. Similar to J48 and PART, errors occur particularly within the densely populated regions of the duration and balance visualization.

Age, job and contact type do not individually produce a clear separation between yes and no. This supports the need for a model to consider combinations of attributes rather than relying on any single demographic characteristic.

Logistic Regression incorrectly classified 353 subscribers as no. This was better than J48's 361 false negatives but slightly worse than PART's 342.

Comparison of Misclassification Patterns

Across all three classifiers, the most important common pattern is that the models are much more successful at identifying non-subscribers than subscribers.

J48 correctly identifies 97.1% of no instances but only 30.7% of yes instances. PART reduces the difference slightly by identifying 34.4% of subscribers, while Logistic Regression identifies 32.2%.

The error visualizations also show substantial overlap between the classes across age, job, balance, duration, and contact type. No single demographic attribute clearly separates subscribers from non-subscribers.

Duration appears particularly important because it provides useful differentiation across several plots and forms the root of the J48 decision tree. Nevertheless, the overlapping instances demonstrate that duration alone is not sufficient for accurate subscription prediction.

Possible Preprocessing Improvements

One possible improvement is to address the class imbalance. The dataset contains 4,000 no instances but only 521 yes instances, meaning approximately 88.5% of the dataset belongs to the no class. This class imbalance may contribute to all three classifiers favoring predictions of no.

Techniques such as oversampling the minority yes class or undersampling the majority no class could be investigated. This may improve the classifiers' ability to learn patterns associated with subscribers, although undersampling may result in useful majority-class information being discarded.

A second possible improvement is binning selected continuous attributes, particularly duration. The error visualizations show considerable overlap within densely populated duration ranges. Creating meaningful duration categories such as short, medium, and long calls could potentially allow J48 and PART to produce simpler decision boundaries and rules.

Another option is feature engineering. Since the results suggest that individual attributes such as age or job do not clearly separate the classes, combinations of attributes could potentially provide additional information. For example, interaction-based features involving campaign characteristics, contact type, or duration could be investigated.

2.5 Random Forest (Ensemble Method):

Random Forest was applied to the same dataset using 10-fold cross-validation. It correctly classified 4,026 of the 4,521 instances, producing an overall accuracy of 89.0511%.

For $y = \text{no}$, Random Forest achieved:

- Precision = 0.903
- Recall = 0.982
- F1-score = 0.941

For y = yes, Random Forest achieved:

- Precision = 0.576
- Recall = 0.188
- F1-score = 0.284

The confusion matrix was:

Actual Class	Predicted (Yes)	Predicted (No)
Yes	98	423
No	72	3928

Random Forest therefore correctly identified only 98 of the 521 subscribers, while 423 actual subscribers were classified as no.

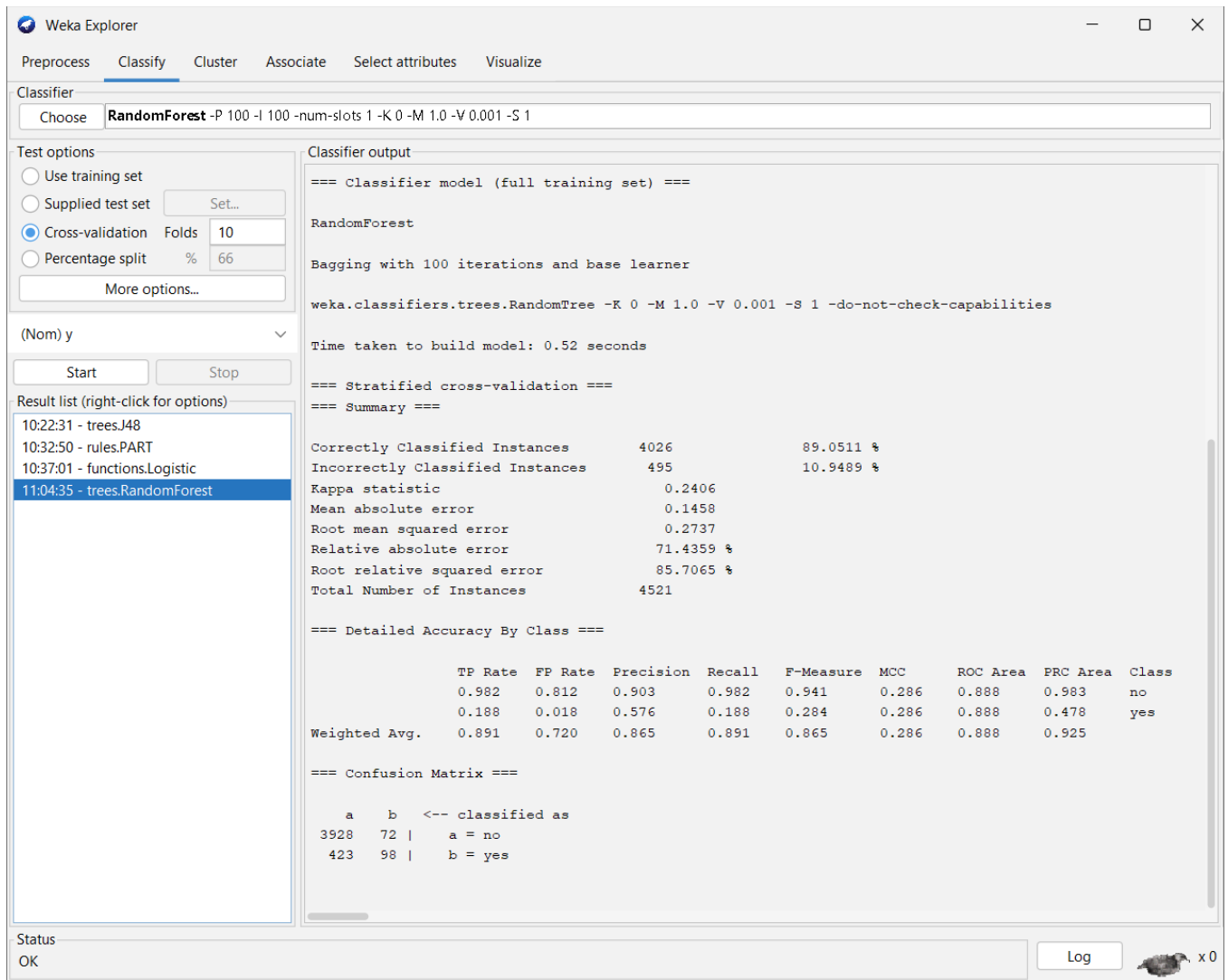


Figure 4 Random Forest Classification Results

Comparison of All Four Models

The following table compares the performance of all four classifiers using overall accuracy and class-specific precision, recall and F1-score for both the no and yes classes. This provides a more complete comparison of each model's ability to classify both non-subscribers and subscribers.

Classifier	Class	Accuracy	Precision	Recall	F1-score
J48	No	89.43%	0.915	0.971	0.942
J48	Yes	89.43%	0.578	0.307	0.401
PART	No	88.34%	0.918	0.954	0.935
PART	Yes	88.34%	0.492	0.344	0.405
Logistic Regression	No	90.22%	0.917	0.978	0.947

Logistic Regression	Yes	90.22%	0.654	0.322	0.432
Random Forest	No	89.05%	0.903	0.982	0.941
Random Forest	Yes	89.05%	0.576	0.188	0.284

Random Forest Comparison

Random Forest did not improve overall performance over the individual classifiers in this experiment. Its accuracy of 89.0511% was lower than Logistic Regression (90.2234%) and J48 (89.4271%), although slightly higher than PART (88.3433%).

More importantly, Random Forest produced the lowest recall for the yes class at 0.188. This means it identified only 18.8% of customers who actually subscribed. Its yes F1-score of 0.284 was also substantially below J48 (0.401), PART (0.405) and Logistic Regression (0.432).

Random Forest achieved a relatively high recall of 0.982 for no, indicating that it was extremely successful at identifying non-subscribers. However, this performance came at the cost of incorrectly classifying 423 of the 521 subscribers as no.

Therefore, the ensemble method did not provide an improvement for the important minority subscription class.

Of the four classifiers, Logistic Regression produced the highest overall accuracy (90.2234%), highest yes precision (0.654), and highest yes F1-score (0.432). PART achieved the highest yes recall (0.344), meaning it detected the greatest proportion of actual subscribers.

Consequently, Logistic Regression provides the strongest overall balance among the four classifiers in these results, while PART may be preferable if the bank places greater importance on identifying as many potential subscribers as possible, even at the cost of more false-positive predictions.

3.0 Clustering and Analysis

3.1 Applying k-Means Clustering:

We tested $k = 2, 3, 4$, and 5 .

K	WCSS	Drops
2	15038	-
3	13764	-1274
4	13025	-739
5	12666	-359

We chose $k = 3$ because it represents the most significant reduction in WCSS (-1274). Beyond this point, the SSE improvement diminishes substantially.

Cluster sizes at $k = 3$ were 1,151 (25%), 2,101 (46%), and 1,269 (28%), which we consider highly balanced because every segment represents a substantial portion of the dataset. None of the clusters are too small to be statistically viable, ensuring that any targeted marketing strategies developed for these profiles will have a meaningful reach.

In terms of interpretability, the clusters at $k = 3$ appear to be driven mainly by age, housing status, education, job type, and call duration, representing profiles such as:

Cluster 0: They work in management, do not have housing loans, and notably engaged in the longest calls (average ~ 325 seconds).

Cluster 1: This is a younger demographic working primarily in blue-collar jobs, with lower education levels (secondary) and active housing loans. They also carry the lowest average account balance.

Cluster 2: A younger demographic with tertiary education working in management. They do not have housing loans and hold the highest average balances, though they spent the least amount of time on calls.

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 5 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

☒ Use training set

☐ Supplied test set Set...

☐ Percentage split % 66

☐ Classes to clusters evaluation (Nom) y v

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

07:38:13 - SimpleKMeans

07:39:22 - SimpleKMeans

07:45:08 - SimpleKMeans

07:46:30 - SimpleKMeans

Clusterer output

=== Clustering model (full training set) ===

kMeans

====

Number of iterations: 10

Within cluster sum of squared errors: 15037.954970833274

Initial starting points (random):

Cluster 0: 0.133193,Middle-aged,self-employed,divorced,tertiary,no,no,cellular,feb,481,2,unknown,yes

Cluster 1: 0.057878,Middle-aged,technician,married,secondary,yes,no,cellular,nov,215,6,failure,no

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (4521.0)	Cluster# 0 (1767.0)	Cluster# 1 (2754.0)
filtered-0-balance	0.0634	0.0601	0.0597
unfiltered-age	Young	Young	Young
unfiltered-job	management	management	blue-collar
unfiltered-marital	married	married	married
unfiltered-education	secondary	tertiary	secondary
unfiltered-housing	yes	no	yes
unfiltered-loan	no	no	no
unfiltered-contact	cellular	cellular	cellular
unfiltered-month	may	aug	may
unfiltered-duration	263.9613	294.6276	244.2854
unfiltered-campaign	2.7936	3.1822	2.5443
unfiltered-poutcome	unknown	unknown	unknown
y	no	no	no

Time taken to build model (full training data) : 0.07 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	1767 (39%)
1	2754 (61%)

Status

OK

Log x 0

Figure 5 ($K = 2$) K-Means Cluster Results

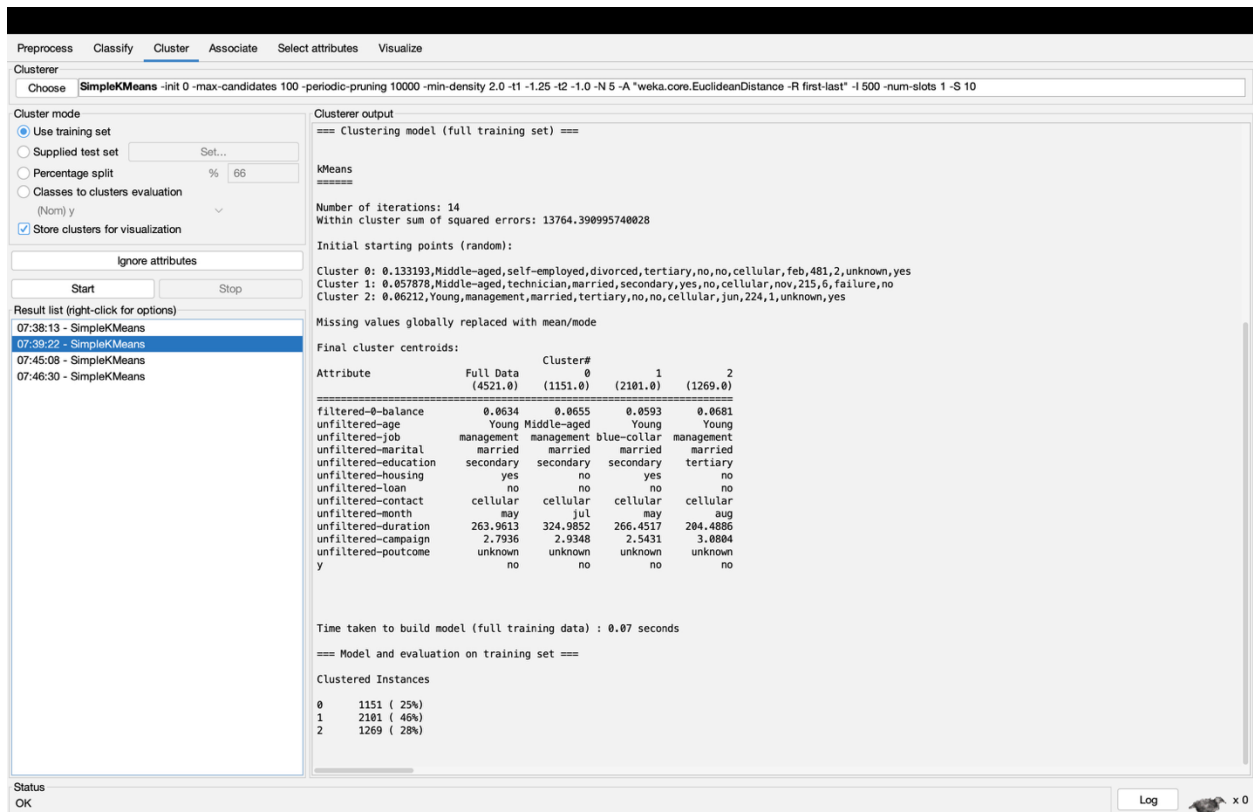


Figure 6 (K = 3) K-Means Cluster Results

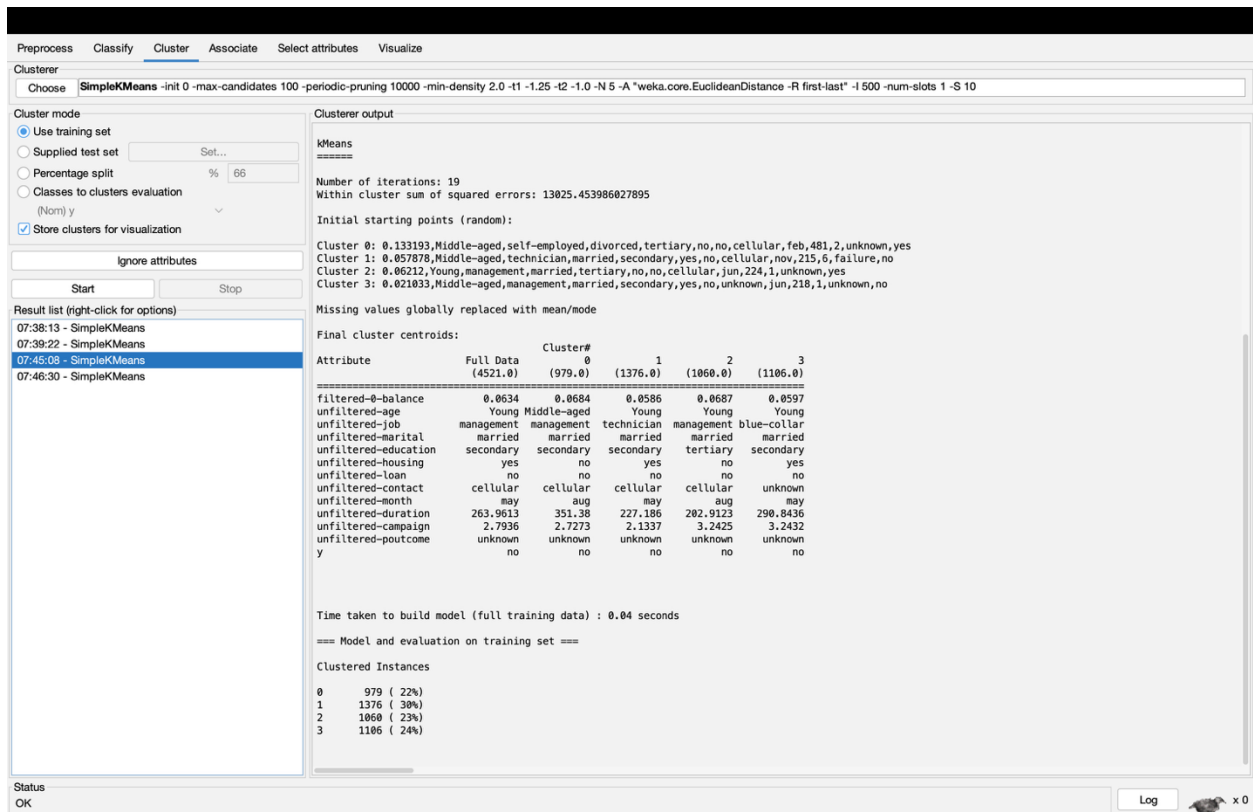


Figure 7 (K = 4) K-Means Cluster Results

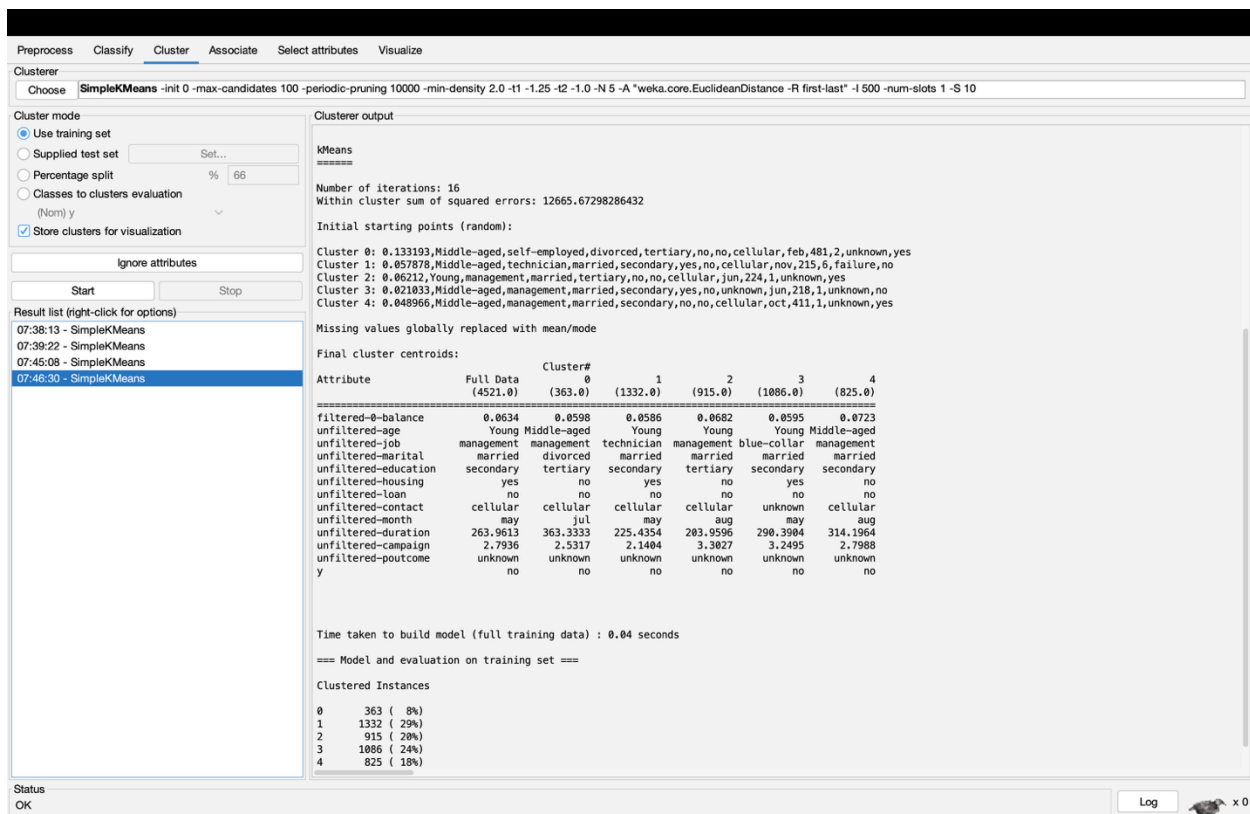


Figure 8 (K = 5) K-Means Cluster Results

3.2 Comparing Clusters with Classification Results:

Subscriber vs non-subscriber

Cluster	Total Instances (N)	Target y="no"	Target y="yes"	Cluster Subscription Rate
Cluster 0	586	456	130	22.18%
Cluster 1	2,044	1,928	116	5.67%
Cluster 2	1,891	1,616	275	14.54%
Total Dataset	4,521	4,000 (88.48%)	521 (11.52%)	11.52% (Base Rate)

Non-subscribers make up the majority of the three clusters with it taking up 77.82%, 94.33% and 85.46% for each of the clusters. This totalled 2318 incorrectly clustered instances, a 51.27 error rate.

Comparative Analysis

Model / Approach	Overall Accuracy	Target y="yes" Precision	Target y="yes" Recall
Logistic Regression	90.22%	0.654	0.322
J48 Decision Tree	89.43%	0.578	0.307

Random Forest	89.05%	0.576	0.188
PART Rules	88.34%	0.492	0.344
k-Means (k=3)	48.73%	0.145	0.528

Discrepancies between clustering and classification

The supervised classifiers use target labels during training. This allows algorithms to evaluate mathematical split criteria to isolate features like specifically separate “yes” from “no”. As for k-means, it operates without access to y (target variable). It groups instances based on spatial proximity, completely indifferent to whether these groupings separate subscribers from non-subscribers.

Furthermore, classification models use a small subset of high leverage attributes for its prediction. “Poutcome” being the largest driver for Logistic Regression. Clustering in contrast, partitions data in terms of broad, high variance demographic and socio-economic dimensions.

Cluster 0: middle-aged management professionals without housing loans who engage in longer calls, cluster 1 of middle-aged blue-collar workers carrying active housing loans with shorter call durations and cluster 2 is of younger management and secondary/tertiary educated clients without housing loans.

In clustering, dense regions of data exert proportional pull-on cluster centroids. Because non-subscribers outnumber subscribers, every cluster centroid is pulled firmly into space dominated by non-subscribers. Without explicit class weighting, unsupervised k-Means cannot mathematically isolate a minority class (11.52%) into a pure, dedicated cluster.

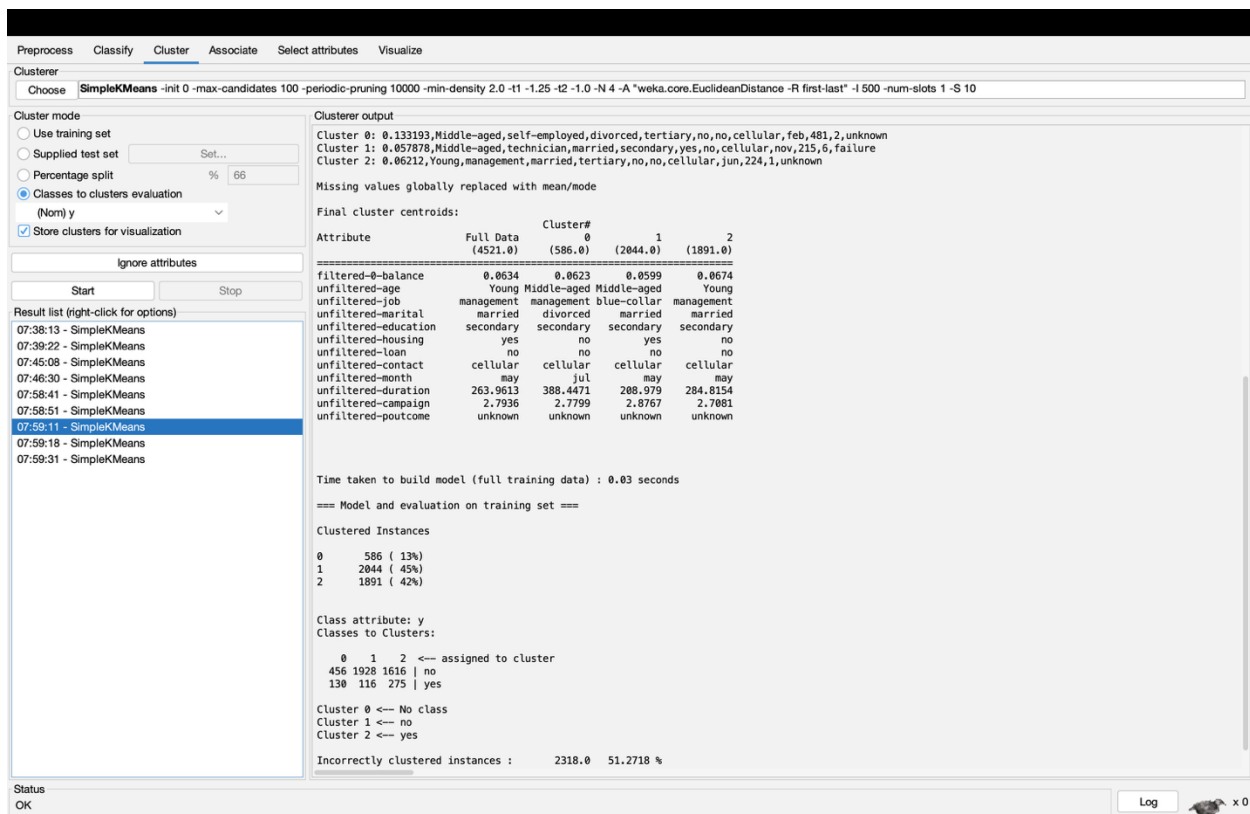


Figure 9 Classes to Cluster ($k=3$)

4.0 Advanced Evaluation Metrics

4.1 Evaluation Using Confusion Matrix:

Logistic Regression		
Actual Class	Predicted (Yes)	Predicted (No)
Yes	168 (TP)	353 (FN)
No	89 (FP)	3,911 (TN)

J48 Decision Tree		
Actual Class	Predicted (Yes)	Predicted (No)
Yes	160 (TP)	361 (FN)
No	117 (FP)	3883 (TN)

PART Decision Tree		
Actual Class	Predicted (Yes)	Predicted (No)
Yes	179 (TP)	342 (FN)
No	185 (FP)	3815 (TN)

Random Forest		
Actual Class	Predicted (Yes)	Predicted (No)
Yes	98 (TP)	423 (FN)
No	72 (FP)	3928 (TN)

Model	Overall Accuracy	Precision	Recall	“Yes” F1-Score	“No” F1-Score	Weighted Avg. F1-Score	ROC Area	PR Area
Logistic Regression	90.22%	0.654	0.322	0.431	0.946	0.887	0.908	0.518
J48 Decision Tree	89.43%	0.578	0.307	0.401	0.942	0.880	0.842	0.435
Random Forest	89.05%	0.576	0.188	0.283	0.941	0.865	0.898	0.491
PART Rules	88.34%	0.492	0.344	0.405	0.935	0.874	0.851	0.419

Critical Thinking

Overall classification accuracy is not an informative and potentially misleading metric for this dataset. Of the dataset, 4000 (88.48%) chose to not subscribe while only 521 (11.52%) subscribed. As there is such an uneven distribution, overall accuracy only reflects the performance on the dominant class rather than the model’s ability to detect the minority class.

A model such as ZeroR whereby every client is classified as $y = \text{"no"}$ due to it being the majority class would have a high accuracy as in the dataset, 4000 (88.48%) chose to not subscribe as such implementing ZeroR would have the accuracy of 88.48%. As for whether this would be useful to the bank, it would not be useful as the recall for ZeroR would be 0 as all clients would be flagged as not subscribing, thus missing out on 521 potential term deposit clients, in turn missing out on revenue.

High Recall ensures the bank minimizes lost revenue opportunities by capturing actual subscribers, Precision prevents wasted telemarketing expenses on cold leads, and the F1-score provides the necessary harmonic balance to evaluate true model effectiveness free from majority-class bias.

Logistic Regression stands out as the most reliable model overall. It offers the highest accuracy, Precision and top F1-score, with PART serving as a viable secondary alternative when maximum subscriber recall is needed more than precision and the F-1 score.

4.2 Practical Recommendations:

J48's root split occurs at duration = 205 seconds, this is supported by Logistic Regression's duration coefficient (-0.0042) that calls falling below this threshold leading almost entirely to non-subscription outcomes. The opposite has a positive correlation with predictions moving away from "no" and toward subscription. Thus the recommendation would be to have call-center staff maintain engagement with clients past the 205 second mark.

PART's first rule states: "outcome = success" as such $y = \text{"yes"}$. Logistic Regression assigns "outcome = success" as the strongest coefficient in the model (-2.2482). With this the recommendation would be to flag clients with "outcome = success" as being a priority for recontact.

k-Means (k=3) profiling shows Cluster 0 (586 clients: middle-aged, management, no housing loan, longest average call duration) converts at 22.18%, while Cluster 1 (2,044 clients: middle-aged, blue-collar, has a housing loan, shortest average call duration) converts at only 5.67%. Therefore, recommendation would be to move focus and call center hours to attend to the clients in cluster 0 rather than cluster 1

Logistic Regression's campaign (number of contacts) coefficient is negative, thus indicating diminishing returns as contact frequency rises. The recommended action would be to limit contact attempts.

5.0 Mark Allocation Sheet

Name	Student ID	Percentage Contribution
Anav Chand	S11221203	100%
Vishant Kumar	S11230430	100%
James Kado	S11200776	100%
Rohan Nandan	S11234883	100%

6.0 Appendix

=== Classifier model (full training set) ===

PART decision list

unfiltered-duration <= 211 AND

unfiltered-poutcome = unknown AND

unfiltered-month = may: no (616.95/1.0)

unfiltered-duration <= 383 AND

unfiltered-poutcome = unknown AND

unfiltered-duration > 75 AND

unfiltered-month = jul: no (437.47/9.0)

unfiltered-duration <= 211 AND

unfiltered-duration <= 75: no (521.3/1.0)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-duration <= 334 AND

unfiltered-contact = unknown: no (417.93/4.5)

unfiltered-duration <= 645 AND

unfiltered-poutcome = failure AND
unfiltered-loan = no AND
unfiltered-duration <= 158: no (112.05/2.68)

unfiltered-duration <= 645 AND
unfiltered-poutcome = unknown AND
unfiltered-duration <= 211 AND
unfiltered-month = aug: no (263.97/3.35)

unfiltered-duration <= 645 AND
unfiltered-poutcome = unknown AND
unfiltered-month = nov: no (202.72/11.51)

unfiltered-duration <= 645 AND
unfiltered-poutcome = failure AND
unfiltered-loan = no AND
unfiltered-month = may AND
unfiltered-campaign <= 2: no (60.75/4.17)

unfiltered-duration <= 645 AND
unfiltered-poutcome = unknown AND
unfiltered-housing = yes AND
unfiltered-month = may AND
unfiltered-marital = married AND
unfiltered-education = secondary: no (75.38/2.4)

unfiltered-duration <= 645 AND
unfiltered-poutcome = unknown AND

unfiltered-month = jan AND

unfiltered-job = management: no (12.41)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-month = jan AND

unfiltered-job = admin.: no (11.98)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-month = jan AND

unfiltered-job = technician: no (10.77/0.55)

unfiltered-duration <= 645 AND

unfiltered-poutcome = other AND

unfiltered-month = may: no (40.57/1.38)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-month = jul: no (77.62/6.69)

unfiltered-duration <= 645 AND

unfiltered-loan = yes AND

unfiltered-poutcome = failure: no (61.68/1.19)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-age = Young AND

unfiltered-loan = yes: no (55.42/2.56)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-month = jan AND

unfiltered-job = blue-collar: no (8.91)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-month = may AND

unfiltered-age = Middle-aged: no (54.5/2.13)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-age = Young AND

unfiltered-campaign <= 6 AND

unfiltered-education = secondary AND

unfiltered-month = apr: no (50.88/7.7)

unfiltered-duration <= 645 AND

unfiltered-poutcome = unknown AND

unfiltered-age = Young AND

unfiltered-campaign > 6: no (24.71)

unfiltered-duration <= 622 AND

unfiltered-poutcome = unknown AND

unfiltered-month = jan AND

unfiltered-job = unemployed: no (4.91)

unfiltered-duration <= 622 AND
unfiltered-poutcome = unknown AND
unfiltered-month = nov: no (4.67/0.61)

unfiltered-duration <= 622 AND
unfiltered-poutcome = unknown AND
unfiltered-month = jan AND
unfiltered-job = services: no (4.35)

unfiltered-duration <= 622 AND
unfiltered-poutcome = unknown AND
unfiltered-month = may AND
unfiltered-job = blue-collar: no (30.49/2.57)

unfiltered-duration <= 306 AND
unfiltered-poutcome = other: no (69.8/11.59)

unfiltered-duration <= 758 AND
unfiltered-poutcome = unknown AND
unfiltered-duration <= 322 AND
unfiltered-month = aug AND
unfiltered-contact = cellular: no (74.45/5.61)

unfiltered-duration <= 211 AND
unfiltered-poutcome = unknown AND
unfiltered-marital = married: no (106.52/11.25)

unfiltered-duration <= 766 AND
unfiltered-poutcome = failure AND
unfiltered-month = apr: no (31.61/2.29)

unfiltered-duration <= 766 AND
unfiltered-poutcome = unknown AND
unfiltered-month = may AND
unfiltered-loan = no AND
unfiltered-contact = unknown AND
unfiltered-education = tertiary AND
unfiltered-duration <= 735: no (17.97)

unfiltered-poutcome = failure AND
unfiltered-loan = no AND
unfiltered-month = nov: no (27.83/3.18)

unfiltered-poutcome = success AND
unfiltered-housing = no AND
unfiltered-campaign <= 1: yes (46.37/8.34)

unfiltered-duration <= 766 AND
unfiltered-poutcome = unknown AND
unfiltered-month = feb AND
unfiltered-duration <= 465: no (55.62/5.75)

unfiltered-duration <= 766 AND
unfiltered-contact = unknown AND
unfiltered-month = jun: no (54.98/7.4)

unfiltered-duration <= 793 AND
unfiltered-month = jan AND
unfiltered-marital = single: no (16.11/1.0)

unfiltered-duration <= 793 AND
unfiltered-month = apr AND
unfiltered-housing = yes: no (51.99/10.06)

unfiltered-duration <= 793 AND
unfiltered-month = may AND
unfiltered-loan = yes AND
unfiltered-contact = cellular: yes (4.3/0.13)

unfiltered-duration <= 793 AND
unfiltered-month = may AND
unfiltered-poutcome = unknown AND
unfiltered-loan = no AND
unfiltered-job = admin.: no (12.88/2.0)

unfiltered-month = may AND
unfiltered-age = Young AND
unfiltered-contact = unknown AND
unfiltered-marital = married: no (29.2/4.0)

unfiltered-duration > 793 AND
unfiltered-month = aug AND
unfiltered-campaign > 2: yes (14.0)

unfiltered-month = jan AND

unfiltered-marital = married AND

unfiltered-age = Young: no (6.69)

unfiltered-month = aug AND

unfiltered-housing = no AND

unfiltered-contact = cellular AND

unfiltered-marital = married AND

unfiltered-poutcome = unknown AND

unfiltered-age = Middle-aged AND

unfiltered-job = management: no (13.12/1.0)

unfiltered-duration <= 205 AND

unfiltered-month = jun: no (15.03/2.0)

unfiltered-month = mar AND

unfiltered-job = management AND

unfiltered-housing = no: no (4.76/1.88)

unfiltered-month = mar AND

unfiltered-job = admin.: yes (4.22)

unfiltered-month = may AND

unfiltered-duration <= 724 AND

unfiltered-job = technician: no (20.55/4.52)

unfiltered-month = aug AND

unfiltered-housing = no AND

unfiltered-contact = cellular AND

unfiltered-poutcome = failure: no (9.02/1.15)

unfiltered-month = oct AND

unfiltered-job = technician: yes (7.06/2.12)

unfiltered-month = nov AND

unfiltered-contact = cellular AND

unfiltered-age = Middle-aged AND

unfiltered-housing = yes AND

unfiltered-duration <= 931: no (6.31/1.0)

unfiltered-poutcome = success: yes (59.21/24.56)

unfiltered-duration <= 238 AND

unfiltered-job = management AND

unfiltered-housing = no: no (16.52/3.37)

unfiltered-month = oct AND

unfiltered-job = management: yes (7.5/1.11)

unfiltered-month = sep AND

unfiltered-education = tertiary AND

unfiltered-housing = no AND

unfiltered-age = Young: no (7.67/2.58)

unfiltered-month = dec AND

unfiltered-duration <= 402: no (4.21/1.0)

unfiltered-month = oct AND

unfiltered-job = admin.: yes (5.88/1.41)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-poutcome = unknown AND

unfiltered-job = retired: no (6.81/1.08)

unfiltered-month = may AND

unfiltered-duration <= 724 AND

unfiltered-job = management AND

unfiltered-poutcome = unknown: no (14.94/3.42)

unfiltered-job = blue-collar AND

unfiltered-month = feb: no (11.78/1.55)

unfiltered-month = sep AND

unfiltered-education = secondary: no (4.35/0.15)

unfiltered-month = sep AND

unfiltered-education = tertiary: yes (4.31/0.25)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-poutcome = unknown AND

unfiltered-job = admin.: no (5.58/1.24)

unfiltered-month = aug AND

unfiltered-poutcome = failure: yes (5.75/1.54)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-job = management AND

unfiltered-education = tertiary AND

unfiltered-housing = no AND

unfiltered-campaign <= 2: no (6.2/1.0)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-poutcome = unknown AND

unfiltered-job = blue-collar AND

unfiltered-duration > 490: yes (4.47/1.36)

unfiltered-job = retired AND

unfiltered-poutcome = unknown AND

unfiltered-month = jun: yes (6.48/0.91)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-poutcome = unknown AND

unfiltered-job = management AND

unfiltered-education = tertiary AND

unfiltered-housing = no: yes (6.4/2.4)

unfiltered-month = aug AND

unfiltered-contact = cellular AND

unfiltered-job = management: no (7.88/1.32)

unfiltered-job = management AND

unfiltered-education = tertiary AND

unfiltered-age = Middle-aged: yes (18.72/3.95)

unfiltered-month = aug AND

unfiltered-job = blue-collar: no (5.28/0.06)

unfiltered-month = may AND

unfiltered-job = student: no (7.39/1.2)

unfiltered-job = retired AND

unfiltered-poutcome = unknown: yes (28.11/9.63)

unfiltered-month = jan AND

unfiltered-marital = married AND

unfiltered-campaign > 1: no (5.23/0.37)

unfiltered-poutcome = other: yes (27.18/9.85)

unfiltered-job = student: yes (12.42/4.58)

unfiltered-job = admin. AND

unfiltered-month = may: yes (5.61/0.56)

unfiltered-loan = yes AND

unfiltered-contact = unknown: no (5.59/0.47)

unfiltered-month = nov: no (16.54/6.61)

unfiltered-month = may AND

unfiltered-job = technician: yes (10.64/3.52)

unfiltered-month = may AND

unfiltered-job = blue-collar AND

unfiltered-marital = married AND

unfiltered-education = secondary AND

unfiltered-duration <= 814: no (5.9)

unfiltered-month = may AND

unfiltered-job = services AND

unfiltered-housing = yes AND

unfiltered-duration > 365 AND

unfiltered-campaign <= 3: yes (7.55/0.67)

unfiltered-month = may AND

unfiltered-job = services: no (10.93/0.3)

unfiltered-month = sep: no (3.17/1.13)

unfiltered-month = may AND

unfiltered-job = entrepreneur AND

unfiltered-age = Young AND

filtered-0-balance > 0.046778: no (2.71/0.9)

unfiltered-month = may AND

unfiltered-job = management AND

unfiltered-contact = unknown: no (4.06/0.8)

unfiltered-job = technician AND

unfiltered-age = Young AND

unfiltered-poutcome = unknown AND

unfiltered-housing = no: no (19.34/4.61)

unfiltered-marital = single AND

unfiltered-job = management: yes (13.37/1.13)

unfiltered-education = tertiary AND

unfiltered-contact = unknown: yes (7.79/2.72)

unfiltered-education = tertiary AND

unfiltered-job = entrepreneur: no (7.82/2.11)

unfiltered-education = tertiary AND

unfiltered-job = management AND

unfiltered-month = jul: no (7.53/1.28)

unfiltered-month = may AND

unfiltered-job = management AND

filtered-0-balance > 0.048791: yes (4.93/0.39)

unfiltered-month = may AND
unfiltered-housing = yes AND
unfiltered-job = management: no (3.62/0.09)

unfiltered-month = may AND
unfiltered-housing = no: no (3.18)

unfiltered-month = jul AND
unfiltered-job = admin.: yes (4.87/1.44)

unfiltered-month = may AND
unfiltered-job = self-employed: yes (2.83/1.15)

unfiltered-month = may AND
unfiltered-job = blue-collar AND
unfiltered-contact = unknown AND
unfiltered-education = secondary AND
unfiltered-duration <= 750: no (2.77)

unfiltered-month = jul AND
unfiltered-job = management: yes (3.25/0.2)

unfiltered-job = management: no (10.51/2.53)

unfiltered-poutcome = failure AND
unfiltered-education = tertiary: no (9.3/1.48)

unfiltered-month = jan AND

unfiltered-housing = no: yes (3.0)

unfiltered-month = oct: yes (8.12/3.08)

unfiltered-month = jan: no (3.0/1.0)

unfiltered-month = jul AND

unfiltered-job = technician: yes (2.19/0.65)

unfiltered-month = aug AND

unfiltered-age = Middle-aged AND

unfiltered-job = services AND

filtered-0-balance > 0.055838: no (4.44/0.12)

unfiltered-month = jul AND

unfiltered-job = entrepreneur AND

unfiltered-duration > 959: yes (2.19)

unfiltered-month = mar AND

filtered-0-balance > 0.048295: no (2.73/0.65)

unfiltered-month = feb AND

unfiltered-housing = yes: no (8.31/2.56)

unfiltered-month = feb AND

unfiltered-age = Middle-aged: yes (6.5/2.05)

unfiltered-month = jun: yes (31.5/14.4)

unfiltered-month = may AND

unfiltered-job = blue-collar AND

unfiltered-marital = married: no (8.27/2.85)

unfiltered-month = feb: no (7.99/3.42)

unfiltered-month = may AND

unfiltered-job = blue-collar: yes (7.33/1.66)

unfiltered-month = may: no (5.36/1.08)

unfiltered-month = apr AND

unfiltered-job = blue-collar AND

unfiltered-marital = married: no (3.8/0.53)

unfiltered-month = apr AND

unfiltered-job = technician: no (3.44/1.26)

unfiltered-month = apr AND

unfiltered-education = tertiary: yes (4.0/1.0)

unfiltered-month = apr AND

unfiltered-education = secondary AND

unfiltered-campaign > 1: yes (3.42/0.89)

unfiltered-month = apr AND

unfiltered-education = secondary AND

filtered-0-balance > 0.051637: no (3.01)

unfiltered-month = apr: yes (5.0/0.53)

unfiltered-month = jul AND

unfiltered-job = services AND

unfiltered-loan = no: yes (3.99/0.8)

unfiltered-month = jul AND

unfiltered-job = blue-collar AND

unfiltered-age = Young: yes (12.0/3.96)

unfiltered-month = jul: no (19.74/7.29)

unfiltered-month = aug AND

unfiltered-job = technician AND

unfiltered-education = secondary AND

unfiltered-marital = married AND

filtered-0-balance > 0.046228: yes (5.23/1.53)

unfiltered-month = aug AND

unfiltered-job = technician AND

unfiltered-marital = married: no (5.9/1.31)

unfiltered-month = aug AND

unfiltered-loan = no AND

unfiltered-job = housemaid: no (3.28/1.06)

unfiltered-month = aug AND

unfiltered-loan = no AND

unfiltered-job = technician AND

unfiltered-marital = single AND

filtered-0-balance ≤ 0.103314 : no (2.81/0.59)

: yes (17.61/4.89)

Number of Rules : 125